

Extrusion-modulated DnaA activity oscillations coordinate DNA replication with biomass growth

Reviewed Preprint

v2 • October 17, 2025

Revised by authors

Reviewed Preprint

v1 • August 6, 2025

Dengjin Li, Hai Zheng, Yang Bai, Zheng Zhang, Hao Cheng, Xiongliang Huang, Ting Wei, Matthew Wook Chang, Arieh Zaritsky, Terence Hwa, Chenli Liu ✉

Key Laboratory of Quantitative Synthetic Biology, Shenzhen Institute of Synthetic Biology, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China • University of Chinese Academy of Sciences, Beijing, China • NUS Synthetic Biology for Clinical and Technological Innovation and Department of Biochemistry, Yong Loo Lin School of Medicine, National University of Singapore, Singapore, Singapore • National Centre for Engineering Biology, Singapore, Singapore • Faculty of Natural Sciences, Ben-Gurion University of the Negev, Be'er Sheva, Israel • Department of Physics & Department of Molecular Biology, University of California, San Diego, La Jolla, United States

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This work provides high-precision single-cell data on the relationship between DnaA activity and cell size, offering **important** insights for the field of cell cycle control. These findings motivate a novel and intriguing hypothesis for DNA replication initiation -the "extrusion model"- in which DNA-binding proteins modulate free DnaA availability in response to biomass-DNA imbalance. While the current indirect evidence does not fully establish the model, an experimental perturbation involving H-NS offers **convincing** support for its plausibility, laying the groundwork for future investigation.

<https://doi.org/10.7554/eLife.107214.2.sa3>

Abstract

Robust control of DNA replication is fundamental to bacterial proliferation. In *Escherichia coli*, replication initiation is thought to be regulated by oscillations in DnaA activity, driven by DnaA-chromosome interactions that differ among leading models. However, direct evidence linking these oscillations to replication initiation has been lacking, and existing models fail to explain the observed decoupling of replication initiation from *dnaA* expression. Here, we establish a direct link between DnaA activity and replication initiation by demonstrating robust oscillations in DnaA activity, which peak precisely at replication initiation across diverse growth conditions and genetic perturbations. Notably, these oscillations persist even when *dnaA* transcription remains constant, suggesting a regulatory mechanism that modulates DnaA activity independently of its expression. Additionally, we propose an extrusion model in which DNA-binding proteins sense biomass-DNA imbalance and extrude DnaA from the chromosome to trigger replication, overcoming limitations of existing models.

Consistent with this model, perturbation of the nucleoid-associated protein H-NS modulates DnaA activity and replication timing, supporting its mechanistic validity.

Introduction

Bacterial proliferation requires precise coordination between DNA replication and biomass growth to maintain cellular homeostasis. While biomass accumulates exponentially, DNA synthesis proceeds in a stepwise manner, creating temporal imbalances that demand continuous correction throughout the cell cycle (**Fig. 1A-B**). Failure to synchronize these processes risks catastrophic outcomes, such as growth defects, incomplete replication, unequal genetic segregation, and cell death (Boye et al., 1996; Khodursky et al., 2015; Lu et al., 1994; Mäkelä et al., 2024; Zaritsky & Pritchard, 1971). Central to this synchronization in *Escherichia coli* is the initiator protein DnaA, a DNA-binding protein that binds specifically to an asymmetric 9-bp DnaA-box, whose consensus sequence is TTWNTNCACA (Hansen & Atlung, 2018; Speck et al., 1999). Although DnaA binding to the replication origin (*oriC*) triggers replication initiation (Dong et al., 2023; Kaguni, 2011; Ohbayashi et al., 2020; Ozaki & Katayama, 2012; Sekimizu et al., 1987), how DnaA activity—defined as the capacity to initiate replication—is dynamically regulated to coordinate DNA synthesis with cellular growth, particularly when DNA content and biomass diverge, remains a fundamental unresolved question in cell cycle biology.

Two competing models have dominated the field. The titration model posits that DnaA activity is governed by the availability of free DnaA, which accumulates until it surpasses the buffering capacity of chromosomal DnaA-boxes (Christensen et al., 1999; Hansen et al., 1991). Here, replication initiation resets the system by increasing DNA content (and thus DnaA-boxes), titrating excess DnaA. In contrast, the switch model emphasizes the ATP/ADP-bound states of DnaA: DnaA-ATP, synthesized during biomass growth, triggers initiation, while hydrolysis to DnaA-ADP following replication resets the cycle (Berger & Wolde, 2022; Donachie & Blakely, 2003; Fu et al., 2023; Hansen & Atlung, 2018; Katayama et al., 2017). Crucially, the titration model depends on ongoing DnaA synthesis to reflect biomass accumulation, whereas the switch model relies on nucleotide-state transitions, independent of total DnaA levels. Despite these mechanistic differences, both models consider that replication initiation is regulated by oscillations in DnaA activity, driven by DnaA-chromosome interactions (**Fig. 1C**). However, direct evidence linking these oscillations to replication initiation remains lacking. Moreover, neither model fully explains recent experimental observations: replication initiation persists for multiple generations even when *dnaA* transcription is completely inhibited (Knöppel et al., 2023). This discrepancy suggests an uncharacterized mechanism that senses biomass-DNA imbalance and directly modulates DnaA activity.

Here, using a synthetic reporter coupled with single-cell mRNA fluorescence *in situ* hybridization (FISH), we demonstrate that DnaA activity—governed by free DnaA concentration, DnaA-ATP/ADP ratio, and orisome assembly competence—peaks precisely at initiation across diverse growth conditions and genetic backgrounds, even when *dnaA* transcription is held constant. DnaA activity oscillations persist independently of transcriptional feedback, pointing to a post-translational regulatory mechanism. We propose an extrusion model in which nucleoid-associated proteins (NAPs) sense biomass-DNA imbalance and dynamically extrude DnaA from the chromosome, liberating it to activate *oriC*. This model not only explains the decoupling of replication timing from *dnaA* expression but also integrates disparate experimental findings. Perturbations of the NAP, H-NS for instance, modulate DnaA activity and replication timing, directly supporting the model's predictions. By unifying regulatory principles across transcription, protein activity, and chromosome dynamics, our findings advance the paradigm for bacterial replication control.

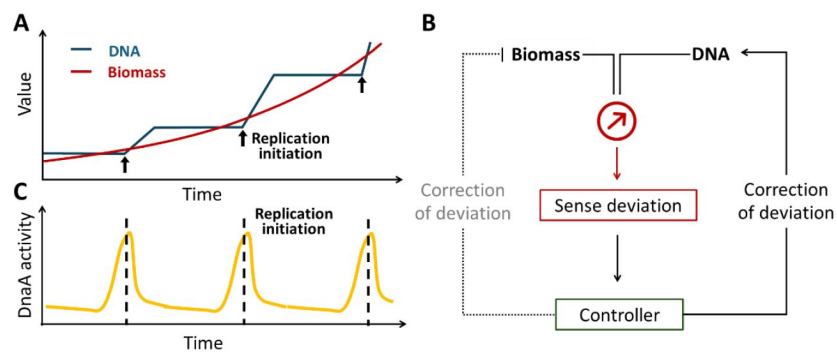


Figure 1.

Coordination of biomass growth and DNA replication through replication initiation and DnaA activity oscillations.

(A) Schematic representation of the discrepancy between biomass accumulation and DNA replication. While biomass grows exponentially, DNA synthesis progresses linearly, necessitating replication initiation events to maintain coordination. **(B)** Mechanistic model for biomass-DNA coordination in bacteria. A molecular sensor detects deviations between cell mass and DNA content, transmitting this information to regulatory controllers that compensate by either increasing DNA replication or restricting biomass accumulation. **(C)** Illustration of cyclic DnaA activity oscillations aligning with replication initiation to ensure precise cell cycle control.

Results

Development of a DnaA activity reporter system

It has been known that changing DnaA expression levels affects DnaA activity, and further changes initiation mass (Berger & Wolde, 2022; Løbner-Olesen et al., 1989; Zheng et al., 2020). To investigate this further, we engineered a *dnaA*-titratable strain, where *dnaA* transcription is controlled by an inducible P_{tet} promoter (Lutz & Bujard, 1997) (Fig. 2A). This system enabled precise modulation of *dnaA* mRNA levels (0.25 to 6 times the wild-type level) via anhydrotetracycline (aTc) induction (0.75–20 ng·ml⁻¹) without altering growth rate or cell size (Fig. 2B–D). As the DnaA expression level increases, DnaA activity reaches the initiation threshold earlier. Given that cell mass remained nearly unchanged, this earlier initiation led to an increase in population-averaged cellular *oriC* numbers (Fig. 2E). Concurrently, the initiation mass was reduced by 50%, and the period from initiation to division ($C + D$) was increased by ~60% (Fig. 2F). Notably, the observed relationship between DnaA expression and initiation mass deviated significantly from both the titration and switch models (Fig. 2G; Supplementary Note S1), suggesting unaccounted regulatory mechanisms.

To dissect DnaA activity dynamics, we constructed a library of synthetic promoters ($n = 67$) by replacing *tetO* operators in P_{tet} with DnaA-boxes (Fig. S1). To assess the degree of DnaA activity-mediated repression, we measured the fold change in GFP expression driven by each synthetic promoter under low and high levels of *dnaA* induction (Fig. 3A). Six synthetic promoters exhibited over 8-fold DnaA-mediated repression, whereas the promoter lacking DnaA-boxes (P_{con}) showed none, and the well-characterized native *dnaA* promoter (P_{native}) (Braun et al., 1985; Saggiaro et al., 2013; Speck et al., 1999) displayed approximately two-fold repression (Fig. 3B; Fig. S1). P_{syn66} , which contains three strong and one weak DnaA-box, exhibited >8-fold repression under high DnaA induction (Fig. 3B). P_{syn66} 's design enables responsiveness to both free DnaA levels (via strong boxes) and the DnaA-ATP/DnaA-ADP ratio (via differential binding to the weak box) (Grimwade & Leonard, 2021; Katayama et al., 2017; Speck et al., 1999). Dose-response assays confirmed that P_{syn66} was progressively repressed across a 40-fold range of *dnaA* expression (Fig. 3C), with no interference from SeqA-mediated sequestration (Fig. 3D–E). These results validated P_{syn66} as a sensitive and specific reporter for DnaA activity.

Decoupling DnaA activity oscillations from *dnaA* transcription fluctuations

To investigate DnaA activity dynamics throughout the cell cycle, we linked P_{syn66} activity to cell size, a surrogate for cell-cycle progression, using single-cell mRNA FISH (Skinner et al., 2013) to quantify *lacZ* mRNA levels in MG1655Δ*lac* cells harboring the P_{syn66} -*lacZ* plasmid (Fig. 4A). A control strain harboring a promoter-less *lacZ* construct (P_{neg}) showed undetectable fluorescence signals (Fig. 4A), confirming the specificity of *lacZ* mRNA detection. Quantitative analysis revealed that P_{syn66} -driven mRNA levels fluctuated approximately 3-fold over the cell cycle, whereas the DnaA-unresponsive constitutive promoter (P_{con}) exhibited stable expression (Fig. 4B). However, fluctuations in mRNA concentration may not directly correspond to changes in DnaA activity due to global factors such as plasmid copy number and RNA polymerase concentration (Balakrishnan et al., 2022; Bintu et al., 2005a; Bintu et al., 2005b). To account for these variables, we used a P_{con} -*lacZ* plasmid to estimate global effects on mRNA concentration and calculated DnaA activity by assuming that it is inversely proportional to $k_{syn66} = \frac{[mZ](P_{syn66})}{[mZ](P_{con})}$.

In wild-type cells (MG1655Δ*lac*) harboring the synthetic reporter system (Fig. 4C), fluctuations in DnaA activity, as denoted by k_{syn66}^{-1} , displayed approximately a 3-fold variation across the cell cycle (Fig. 4D), underscoring the substantial cell cycle-dependent oscillations in DnaA activity.

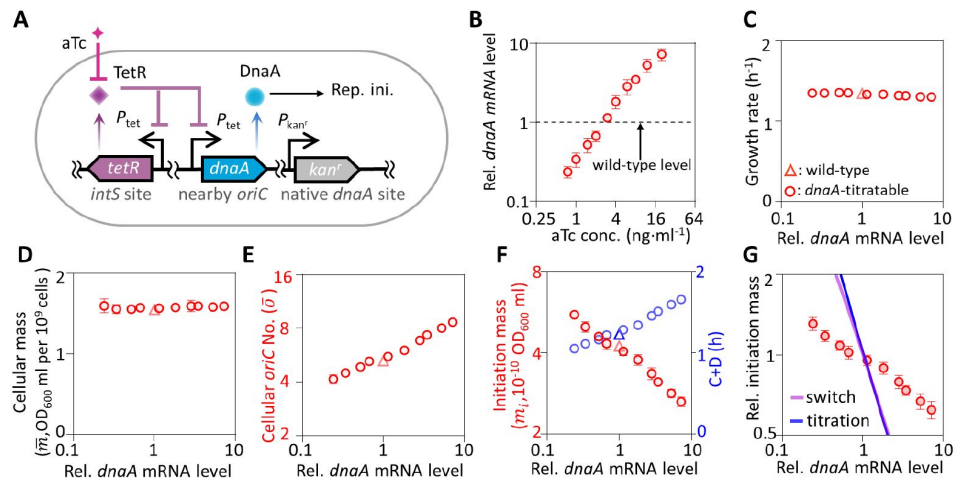


Figure 2.

Construction and characterization of the *dnaA*-titratable strain.

(A) Schematic of the *dnaA*-titratable strain. A *dnaA* gene under the control of the P_{tet} promoter was inserted near *oriC* and regulated by a P_{tet} -*tetR* feedback loop integrated at the *intS* locus, enabling fine-tuned expression control. The native *dnaA* gene was replaced with a kanamycin resistance cassette (*kan^r*). (B–F) Characterization of *dnaA*-titratable cells (circle) and wild-type MG1655 cells (triangle) grown in rich defined medium with glycerol (M6) under varying aTc concentrations. Measured parameters include: (B) *dnaA* mRNA levels; (C) growth rate; (D) population-averaged cellular mass; (E) population-averaged *oriC* numbers; (F) initiation mass (red, left axis) and the initiation to division period (C+D) (blue, right axis). The *dnaA* mRNA levels were normalized to that in wild-type cells. Cellular mass was determined by OD₆₀₀ divided by cell number concentration. *oriC* copy numbers were measured using a run-out assay, and initiation mass was calculated as the ratio of cellular mass to *oriC* numbers. Data represent means ± SD (n = 5 biological replicates). (G) Relationship between relative initiation mass and relative *dnaA* mRNA levels, compared with predictions from the initiation titration model (blue line) and the switch model (purple line). The relative *dnaA* mRNA levels in experiments are compared to relative DnaA expression rate a_A in models. Experimental data are overlaid for validation.

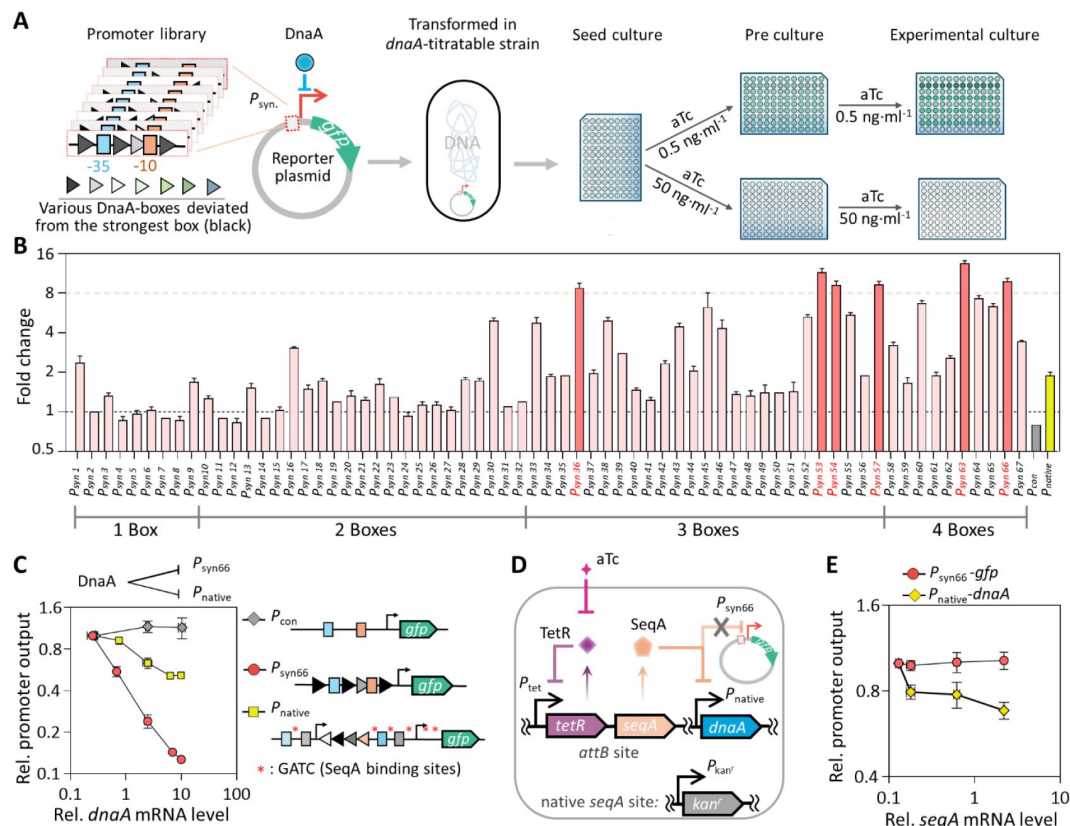


Figure 3.

Development of a DnaA activity reporter system.

(A) Schematic of promoter design and screening. Sixty-seven synthetic promoters were constructed by inserting various DnaA-boxes around the promoter core to drive *gfp* expression in a *dnaA*-titratable strain, where *dnaA* expression was regulated by aTc concentration. After pre-cultivation, GFP fluorescence per OD₆₀₀ was measured using a microplate reader in cells grown under low (0.5 ng·ml⁻¹) or high (50 ng·ml⁻¹) aTc concentrations to determine repression fold-change. **(B)** Repression fold-change of synthetic promoters. P_{con} (a promoter lacking DnaA-boxes) and P_{native} (the endogenous *dnaA* promoter) served as negative and positive controls, shown in gray and yellow, respectively. **(C)** Response curves of three promoters to varying *dnaA* expression levels, with their promoter architectures shown on the right. Promoter activity was assessed by relative *gfp* mRNA levels, normalized to the lowest *dnaA* expression condition. Schematic **(D)** and response curves **(E)** of P_{syn66} and P_{native} responses to SeqA in the *seqA*-titratable strain. Promoter activity was quantified from *gfp* transcript levels in *seqA*-titratable cells containing the P_{syn66}-GFP plasmid, normalized to the lowest *seqA* expression level. All cells were grown in rich defined medium supplemented with glycerol across different aTc concentrations. Data represent mean ± SD from 3 biological replicates **(B,C,E)**.

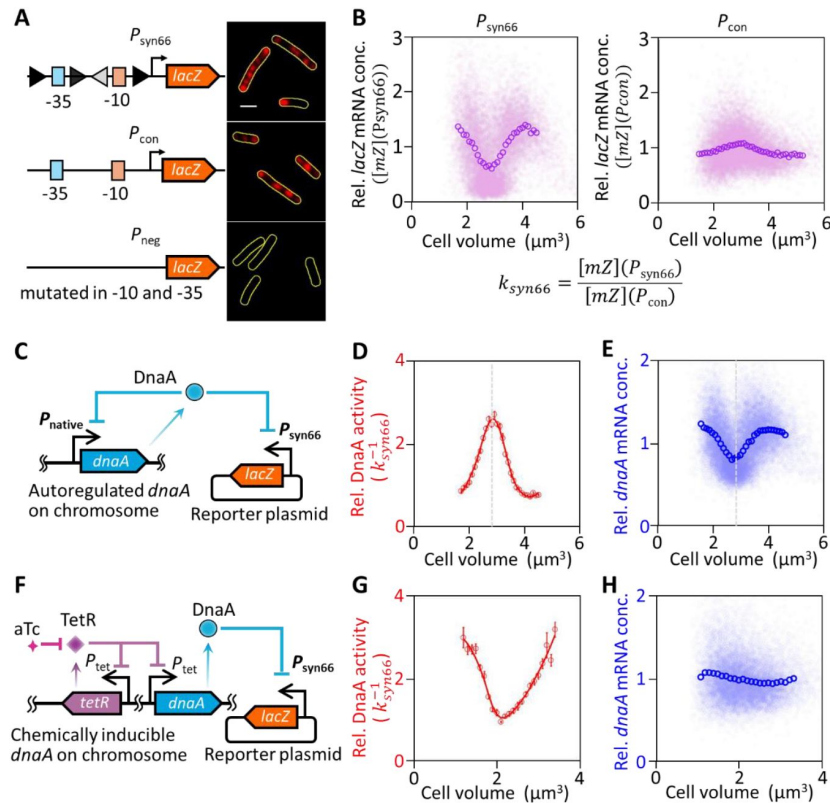


Figure 4.

DnaA activity oscillations decoupled from *dnaA* transcription fluctuations.

(A) Representative *lacZ* mRNA FISH images of MG1655 Δlac cells transformed with *lacZ* expression plasmids driven by P_{syn66} (DnaA-boxes around promoter core), P_{con} (no DnaA-box around promoter core), or P_{neg} (mutated promoter core). Yellow outlines indicate cell boundaries identified from phase-contrast images. **(B)** Relative *lacZ* mRNA concentrations driven by P_{syn66} (left) and P_{con} (right) across different cell volumes. Relative concentrations were determined from volume-specific *lacZ* mRNA fluorescence intensities, normalized to the population average. Volume-binned data for P_{syn66} ($[mZ](P_{syn66})$) and P_{con} ($[mZ](P_{con})$) are shown as open circles and were used to calculate k_{syn66} . **(C)** Schematic of a strain with autoregulated *dnaA* transcription carrying a DnaA activity reporter plasmid. Cell cycle-dependent fluctuations in relative DnaA activity (**D**) and relative *dnaA* mRNA concentrations (**E**) in cells from panel **C**, grown in rich defined medium supplemented with glucose. Relative DnaA activity (k_{syn66}^{-1}), calculated from volume-binned data in panel **B**, was smoothed and plotted as a red curve (**D**). Relative *dnaA* mRNA concentrations were determined from volume-specific *dnaA* mRNA fluorescence intensities, normalized to the population average, with volume-binned data shown as open circles (**E**). Dashed lines indicate the cell volume at peak DnaA activity (**D**) and the minimum *dnaA* mRNA content (**E**). **(F-H)** Same as panels **C-E**, but for cells with aTc induced *dnaA* transcription, grown in rich defined medium supplemented with glycerol under 2 ng·ml⁻¹ aTc induction. More than 8,000 cells were analyzed per growth condition, and all error bars correspond to standard error of the mean (SEM).

We then examined whether the observed fluctuations in DnaA activity were linked to oscillations in *dnaA* transcription. Comparing the dynamics of *dnaA* mRNA and DnaA activity, we found that the peak in DnaA activity coincided with the trough in *dnaA* mRNA levels (dashed line in **Fig. 4D–E**). To test whether *dnaA* transcription drives these oscillations, we eliminated transcriptional fluctuations using the *dnaA*-titratable strain. Despite constant *dnaA* mRNA levels (2 ng·ml⁻¹ aTc induction), DnaA activity was retained consistent oscillations (**Fig. 4F–H**), demonstrating the decoupling of DnaA activity oscillations from transcriptional fluctuations. Our data suggest that post-translational regulation, not transcriptional feedback, governs the DnaA activity.

DnaA activity oscillations are tightly coupled to replication timing

To explore the relationship between DnaA activity oscillations and DNA replication initiation, we examined cells grown under various nutrient conditions supporting different doubling times (30–66 min). Significant fluctuations in DnaA activity were observed across all conditions (**Fig. 5A**). We also estimated the cell volume at replication initiation (V_i) based on population-averaged cell volume and *oriC* number (Bremer et al., 1979; Si et al., 2017; Zheng et al., 2016) (**Fig. S2 A–C**), marking this on the DnaA activity profiles. In a representative birth-to-division cell cycle (Pountain et al., 2024) (**Fig. S2D**), DnaA activity peaks consistently coincided with V_i , indicating a close correlation between DnaA activity and replication initiation (**Fig. 5A**).

Further genetic perturbations, such as titrating DnaA expression, shifted DnaA activity fluctuations, yet the relationship between V^* (the cell volume at maximal DnaA activity in a representative birth-to-division cell cycle) and V_i remained locked (**Fig. 5B**). To quantify this relationship, we compared V_i with V^* , revealing a strictly proportional relationship through the origin (slope = 1.0, $R^2 = 0.98$), indicating equivalence between V^* and V_i (**Fig. 5C**). These results confirm that DnaA activity oscillations are tightly correlated with replication initiation, independent of transcriptional or nutrient perturbations.

An extrusion model reconciles paradoxical initiation events

Very recently, Knöppel *et al.* reported that multiple rounds of initiation occurred even after DnaA synthesis is shut down (Knöppel et al., 2023). In their experiment, DnaA was supplied in excess before expression ceased (Knöppel et al., 2023), potentially allowing continued replication initiation due to residual DnaA. To address this issue, we shut down DnaA expression from its native level using the deactivated miniature Un1Cas12f1 (dUn1Cas12f1) system, guided by sgRNA to repress *dnaA* transcription (**Fig. 6A**). This system effectively diminished *dnaA* transcription to an extremely low level, yet total *oriC* numbers increased fourfold within 90 min, consistent with two rounds of replication initiation (**Fig. 6B**). These findings demonstrate that halting DnaA synthesis does not immediately abolish replication initiation.

Both the titration and switch models predict a close relationship between DnaA activity oscillations and replication initiation (Berger & Wolde, 2022). However, they fail to account for the multiple rounds of initiation observed after DnaA synthesis was shut down. According to the titration model, initiation should cease immediately once DnaA expression is shut down, as the reduction in DnaA concentration would prevent it from reaching the threshold required to trigger new rounds of replication. In contrast, the switch model predicts infinite initiations following the cessation of DnaA expression (**Fig. 6C**). This is because the cessation of DnaA synthesis only reduce the production rate of DnaA-ATP and had no effect on the conversion rates between DnaA-ATP and DnaA-ADP. As a result, the ratio of DnaA-ATP to DnaA-ADP would temporarily decrease, causing a brief lag before initiation resumes indefinitely (**Fig. 6C**).

The failure of both models prompts us to revisit the regulatory mechanisms of replication initiation. In the titration model, a substantial fraction of DnaA proteins binds to DnaA-boxes along the chromosome, raising the possibility that these bound proteins serve as a reservoir of free DnaA, sustaining replication initiation even after DnaA synthesis ceases. We thus considered the

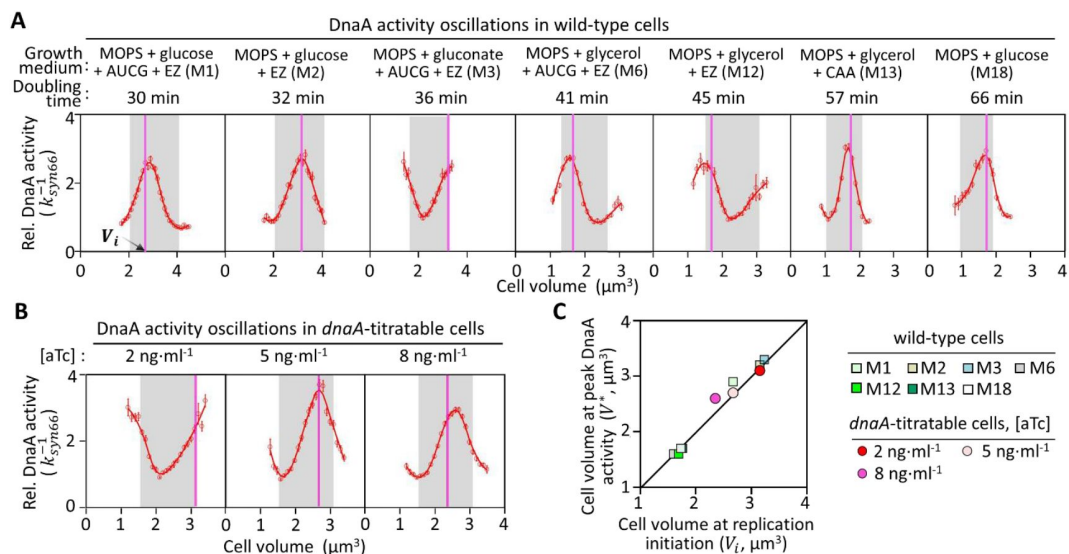


Figure 5.

Tight correlation between DnaA activity oscillations and DNA replication initiation.

(A) Cell cycle-dependent DnaA activity oscillations in wild-type cells across various growth conditions. DnaA activity is represented by k_{syn66}^{-1} . Volume-binned DnaA activity (red circles) was smoothed and plotted as a red curve. The representative birth-to-division cell cycle, defined as the cell volume doubling interval containing the majority of cells, is shaded in gray. The vertical line indicates the cell volume at replication initiation (V_i). **(B)** Cell cycle-dependent DnaA activity oscillations in *dnaA*-titratable cells grown in M6 medium with varying aTc concentrations. More than 8,000 cells were analyzed per growth condition, and error bars show the mean $\pm$ SEM. **(C)** Correlation between V^* and V_i in wild-type (squares) and *dnaA*-titratable (circles) cells. V^* represents the cell volume at the peak of DnaA activity within the representative birth-to-division cell cycle. The black line indicates equivalence between V^* and V_i .

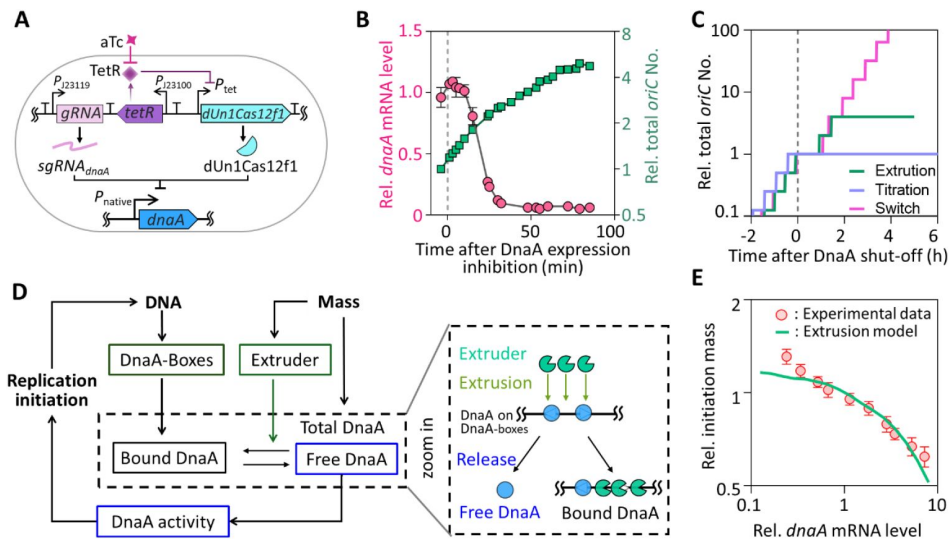


Figure 6.

An extrusion model explains DnaA shut-down dynamics.

(A) Genetic circuit of the deactivated CRISPR-Cas system for *dnaA* transcription shut-down. *dnaA* gene is targeted by a constitutively expressed sgRNA, while dUn1Cas12f1 expression is inhibited by TetR repressor. These transcription units are separated by terminators. The cassette was integrated into the chromosome near the *oriC* locus. DnaA shutdown is induced by the addition of aTc. (B) Time-course of relative *dnaA* mRNA levels (red line, left axis) and total *oriC* number (green line, right axis) following the addition of 50 ng·ml⁻¹ aTc at time 0 (dashed line). *dnaA* mRNA levels were normalized to wild-type levels, and *oriC* numbers were normalized to their initial values. Error bars indicate mean ± SD (n = 3 biologically independent experiments). (C) Predicted increases in total *oriC* number during *dnaA* transcription shut-down based on three models: the titration model, switch model, and extrusion model. Shutdown was simulated by setting *dnaA* transcription to zero at time 0 (dashed line). (D) Schematic of the extrusion model. The model introduces extruder(s) as additional regulators of biomass-DNA coordination, complementing the role of DnaA (left). Increased binding of the extruder to DNA promotes the release of DnaA from DnaA-boxes (right). (E) Comparison of the relationship between relative initiation mass and relative *dnaA* mRNA levels from experimental data (Fig. 2F) and predictions of the extrusion model.

possibility of some extruder(s) alongside DnaA to sense biomass-DNA deviation, and flush DnaA proteins away from the DnaA-boxes, thus releasing free DnaA to dynamically restore DnaA activity, even in the absence of ongoing DnaA synthesis (Fig. 6D [↗](#)).

In this proposed “extrusion model”, the extruder is constitutively expressed, synthesized at a constant rate throughout the cell cycle, accumulating in proportion to biomass growth, enabling replication initiation when free DnaA concentrations exceed the critical threshold, as described in the titration model (Hansen & Atlung, 2018 [↗](#); Hansen et al., 1991 [↗](#)) (Fig. 6D [↗](#)). This model guarantees the long-term coordination of biomass growth and *oriC* abundance, as well as stable oscillations in free DnaA and *oriC* concentrations (Fig. S3 A [↗](#)). Notably, the extrusion model accounts for the persistence of replication initiation and free DnaA oscillations over multiple cell cycles after DnaA synthesis is halted (Fig. 6C [↗](#); Fig. S3 B [↗](#)). Moreover, since the extruder mitigates the impact of perturbations on DnaA synthesis, the model predicts a quantitative relationship between initiation mass and DnaA expression levels that aligns with experimental observations (Fig. 6E [↗](#); Supplementary Note S1). Furthermore, a stochastic implementation of the extrusion model shows the lack of correlation between successive replication initiation events (Fig. S3 C [↗](#)), as observed in recent studies (Si et al., 2019 [↗](#); Witz et al., 2019 [↗](#)).

Perturbation of H-NS validates the extrusion mechanism

The extrusion model could be validated by changing the expression level of the extruder, which modulates DnaA activity by releasing DnaA from DnaA-boxes. Given their DNA binding properties, NAPs (Dillon & Dorman, 2010 [↗](#)) are likely candidates for this extruder role. Therefore, we selected H-NS (Dorman, 2004 [↗](#)), a major NAP, which is capable of promoting the release of DnaA from bound DnaA-boxes *in vitro* (Fig. S4 [↗](#)), to experimentally assess the validity of the extrusion model. To this end, we constructed a *hns*-titratable strain, incorporating a reporting plasmid to capture changes in DnaA activity across various H-NS expression levels (Fig. 7A [↗](#)). Under steady-state cultivation, increasing aTc concentrations from 0 to 50 ng·ml⁻¹ resulted in *hns* mRNA levels ranging from 0.25 to 5 times the wild-type levels, without affecting growth rate (Fig. S5 A-B [↗](#)). Within this range, DnaA activity increased by approximately 60 % (Fig. 7B [↗](#)), while the initiation mass decreased by 30 % (Fig. 7C [↗](#); calculated with Fig. S5 C-D [↗](#)), in qualitative agreement with the predictions of the extrusion model (Fig. S5 E-F [↗](#)).

Since H-NS is a global transcriptional regulator (Hommais et al., 2001 [↗](#)), we wonder whether its effect on DnaA activity was mediated through *dnaA* transcription. To test this, we measured DnaA activity and *dnaA* mRNA levels during an H-NS shift-up. Upon addition of 50 ng·ml⁻¹ aTc, *hns* levels increased more than 20-fold within a short period, followed by approximately 80 % increase in DnaA activity, with no significant change in *dnaA* mRNA levels (Fig. 7D [↗](#)). Additionally, cellular *oriC* content increased significantly within 30 minutes, while cell mass remained unchanged (Fig. 7E [↗](#)). These results suggest that the rapid increase in DnaA activity and accelerated replication initiation induced by H-NS upregulation is not driven by changes in *dnaA* transcription.

Leveraging the *hns*-titratable strain, we re-examined the tight correlation between DnaA activity oscillations and replication initiation. In two different growth media with varying *hns* levels, the peak of DnaA activity fluctuations (V^*) shifted in concert with the independently measured volume at replication initiation (V_i) (Fig. 7F [↗](#)). Plotting V^* against V_i revealed the same equivalency observed in wild-type and *dnaA*-titratable cells (Fig. 7G [↗](#)). Collectively, these results support the notion that DnaA activity oscillations, modulated by extrusion, coordinate DNA replication and biomass growth.

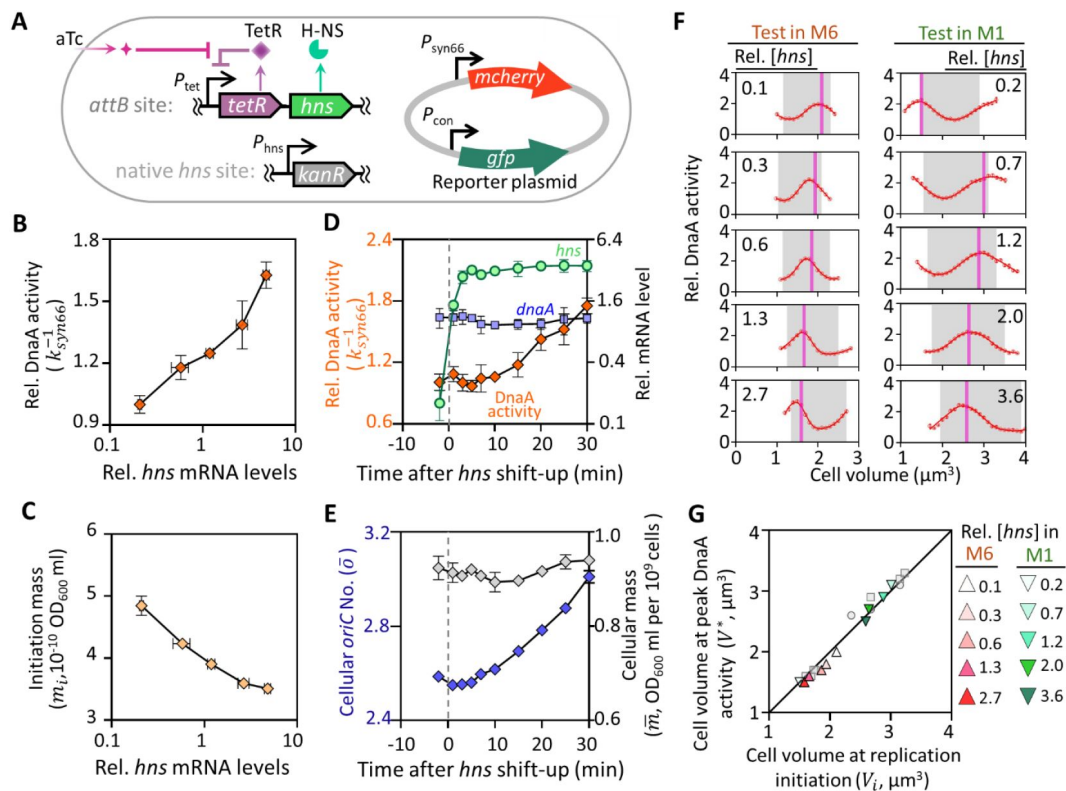


Figure 7.

Titration of *hns* expression modulates DnaA activity and replication initiation.

(**A**) Genetic circuit of the *hns*-titratable strain. Expression of *hns* is controlled by a P_{tet} -*tetR* negative feedback loop integrated at the *attB* site, with the native *hns* coding sequence replaced by a kanamycin resistance gene. The plasmid contains P_{syn66} -*mcherry* and P_{con} -*gfp* expression cassettes was used to assess DnaA activity. DnaA activity (**B**) and initiation mass (**C**) were characterized in M6 medium with varying *hns* expression levels during steady-state cultivation. (**D**) mRNA levels of *hns* (green circles) and *dnaA* (blue squares) relative to wild-type levels, along with DnaA activity (orange rhombus), were measured during *hns* shift-up. *hns* shift-up was induced by the addition of 50 ng·ml⁻¹ aTc at time 0 (dashed line), following steady-state cultivation in M6 medium without aTc. (**E**) Dynamics of population-averaged *oriC* number (blue rhombus, left axis) and cellular mass (gray rhombus, right axis) during *hns* shift-up. Data represent mean $\pm$ SD from 3 biological replicates (**B-E**). (**F**) Cell cycle-dependent DnaA activity oscillations in *hns*-titratable cells cultivated in M1 and M6 media under varying aTc concentrations. Relative *hns* mRNA levels are indicated for each condition. More than 8,000 cells were analyzed for each condition and error bars show the mean $\pm$ SEM. (**G**) Correlation between the volume at maximal DnaA activity (V^*) and the volume at replication initiation (V_i) for *hns*-titratable cells grown in M1 (green down-pointing triangle) and M6 (red up-pointing triangle) media with varying *hns* expression levels. Data for *dnaA*-titratable cells (gray circles) and wild-type *dnaA*-autoregulated cells (**Fig. 5C**) are included for comparison.

Discussion

In this study, we proposed chromosome-driven DnaA activity oscillations as a post-translational “rheostat” coordinating DNA replication with biomass growth in *E. coli*. By developing a SeqA-independent synthetic promoter reporter (P_{syn66}) on a plasmid, we circumvented confounding factors inherent to native systems (Pountain et al., 2024), thereby achieving the direct quantification of DnaA activity dynamics. Crucially, these oscillations persist when *dnaA* transcription is fixed, operate independently of SeqA-mediated repression, and peak precisely at replication initiation across all tested growth conditions. This reveals an essential paradigm: bacteria bypass transcriptional control to synchronize genome duplication with growth through regulation of DnaA-chromosome interactions.

Very recently, Iuliani *et al.* also reported DnaA activity oscillations (Iuliani et al., 2024), which is a significant milestone, innovatively using a chromosomal promoter reporter system and microfluidics for single-cell DnaA activity observations. They revealed key relationships between DnaA activity, cell size, and division. Our work, however, differs in multiple ways. Methodologically, we engineered a *dnaA*-titratable strain and synthetic promoters like P_{syn66} , enabling precise *dnaA* expression control and cleaner DnaA activity reporting. Our mRNA FISH approach uniquely compares DnaA activity and transcription at the single-cell level. Replication initiation volume scales proportionally with peak DnaA activity volume with a slope of 1.0 ($R^2=0.98$, **Fig. 7G**), indicating predictive correspondence rather than absolute coincidence. While population-level V_i estimation cannot resolve single-cell stochasticity, the consistent $V^*: V_i$ relationship across 20 conditions suggests DnaA activity thresholds predict initiation timing within physiological error margins. Regarding research questions, while Iuliani *et al.* focused on DnaA-cell size-division relationships, we addressed the decoupling of replication initiation from *dnaA* expression with our novel extrusion model.

Recently, an integrated “titration-switch” model has been proposed to improve replication cycle stability across different growth rates (Berger & Wolde, 2022; Fu et al., 2023). Although this integrated model predicts a relationship between initiation mass and *dnaA* mRNA levels that aligns closely with steady-state experimental data (**Fig. S6 A**), it fails to account for the additional replication initiation observed after *dnaA* shutdown (**Fig. S6 B**). By incorporating an extruder into this model, we successfully reproduced the additional replication initiation (**Fig. S6 C**), thus validating the extrusion concept and aligning with experimental observations. These findings highlight the critical role of extrusion in refining our mechanistic understanding of biomass-DNA coordination.

Mechanistically, we propose that NAPs sense biomass-DNA imbalances and extrude DnaA from chromosomal sites, converting physical displacement into biochemical activation, a spatial control strategy mirroring eukaryotic chromatin regulation. H-NS emerges as a prime extruder candidate given its DNA-condensing ability (Dame et al., 2000; Zimmerman, 2006) and replication-timing effects (**Fig. 7C**). This functional multiplicity likely enhances robustness, allowing cells to maintain replication homeostasis despite fluctuating environments, an evolutionary advantage paralleling eukaryotic checkpoint system.

These findings refine bacterial cell cycle control as a dynamic interplay between physical chromosome remodeling and metabolic sensing. By tethering DnaA activation to biomass accumulation through extrusion, cells achieve an important capability: noise suppression via spatial filtering of transcriptional fluctuations. Such mechanistic parsimony, using chromosome structure to both store genetic information and regulate replication timing, may represent a

potential organizational strategy. Indeed, parallels to eukaryotic systems are striking: DnaA extrusion resembles ORC licensing through chromatin accessibility (Fragkos et al., 2015 [DOI](#)), while biomass-DNA coupling evokes yeast size-control checkpoints (Xie et al., 2022 [DOI](#)).

While H-NS perturbation supports extrusion mechanism, future work should identify the full extruder interactome and elucidate how metabolic signals modulate their activity. Our synthetic reporter system provides a valuable tool for dissecting the spatial regulation of DNA-binding proteins, with translational potential in antimicrobial development (targeting pathogen-specific extruders) and synthetic biology (engineering tunable replication circuits). More broadly, this work establishes physical chromosome remodeling as a central coordinator of cellular homeostasis, a concept bridging prokaryotic and eukaryotic cell cycle paradigms.

Materials and Methods

Plasmid and strain Construction

Strains used in this study were derived from *E. coli* MG1655. Information for all strains and plasmids are listed in Tables S1 and S2.

Plasmids for reporting DnaA activity

Plasmids expressing GFP, mCherry, or LacZ under synthetic promoters were constructed based on the pPT plasmid backbone (Zong et al., 2017 [DOI](#)). The *lacZ* amplified from the MG1655 genome and mcherry gene amplified from pMD19-*hupA-mcherry* plasmid, were assembled into the BspQI/BsrGI-digested pPT backbone via the One Step Cloning Kit (Vazyme, C112), to generate the pPT-lacZ and pPT-RFP plasmids, respectively. Synthetic promoters, generated by annealing the primer pairs listed in Table S3, were ligated directly into the BsaI-digested pPT, pPT-RFP or pPT-lacZ plasmid as required. The native *dnaA* promoter (P_{native}) amplified from the MG1655 genome was digested with BsaI, and ligated into the BsaI-digested pPT plasmid. The P_{con} -*gfp* cassette amplified from P_{con} -GFP plasmid was assembled into the SpeI-digested P_{syn66} -RFP plasmid to generate the P_{syn66} - P_{con} -FPs plasmid. Primer information for amplifying DNA fragments is detailed in Table S4.

Plasmids for strain construction

For constructing the plasmid pMD19-Rhns, pMD19-RseqA and pMD19-RdnaA, the *hns*, *seqA* and *dnaA* gene were amplified from MG1655 genome, digested, and ligated into the same enzyme-digested pMD19-tetR plasmid. For constructing the plasmid p15A-RdnaA, the T3- P_{tet} -*tetR-dnaA* cassette, *kan^r* gene and p15A-T1 fragment were amplified from plasmid pMD19-RdnaA, pEcCas and PZA31- P_{tet} -M2-GFP, respectively, and then assembled via MultiS One Step Cloning Kit (Vazyme, C113). The plasmid P_CRidnaA1 were obtained by ligating the annealed primer pairs to the BsaI-digested CPP00458 plasmid for the *dnaA* targeting. The chloramphenicol resistance gene *Cm^r* amplified from the PZA31- P_{tet} -M2-GFP plasmid replaced the *kan^r* gene in the pEcCas plasmid via λ Red recombination method, resulting in the CmPcas plasmid. Primer information is detailed in Table S4.

Strain construction

The *seqA*-, and *hns*-titratable strains were constructed as previously described (Zheng et al., 2016 [DOI](#)). Briefly, after transforming pSIM5 plasmid into MG1655 cells, the P_{tet} -*tetR*-driven *seqA* or *hns* cassette was inserted at the *attB* site on the genome, and then the native gene was replaced with the *kan^r* gene amplified from plkm1 plasmid.

For the construction of *dnaA*-titratable strain, we first inserted the P_{tet} -*tetR* cassette into the *intS* site via CRISPR/Cas9- λ Red genome editing system (Jiang et al., 2015) by targeting 20-nucleotide sequence (ttctcgtcagaccagatcc). Then applying the λ Red recombination systems in the CmPcas plasmid, the *bla*: P_{tet} -*dnaA* cassette amplified from the pMD19-RdnaA plasmid via overlap PCR, was inserted between the *gidA* and *gidX* genes, and the negative P_{native} -*dnaA* was replaced with the T₃-Kan-T₁ cassette amplified from the p15A-RdnaA plasmid.

In order to delete the native *lacZ* gene in the genome of the RdnaA1, Rhns1 and MG1655 cells, the homologous recombination fragment amplified from the genome of CL1 strain, was used to knock out the native *lac* operon via CRISPR/Cas9- λ Red editing system (Jiang et al., 2015) by targeting 20-nucleotide sequence (cttcggctcgtatgtgtg).

For the construction of DnaA shut-down strain (CRidnaA1), the P_{J23119} -sgRNA_{*dnaA*}: P_{J23100} -*tetR*: P_{tet} -*dUn1Cas12f1* cassette was amplified from the P_CRidnaA1 plasmid and flanked with homologous sequences to the *asnA* and *viaA* gene. This cassette was then inserted into the MG1655 genome using CRISPR/Cas9- λ Red editing system, targeting the 19-nucleotide sequence (aagccgctgctcagacgc). The CPP00458 plasmid, kindly provided by the Xiongfei Fu lab, contains the Un1Cas12f1 variant, which carries mutations D143R, T147R, D326A, and K330R that abolish its DNA-cleaving activity. However, it retains the ability to bind DNA in a sgRNA-guided manner, thereby inhibiting the expression of the target gene.

All primers for the construction of DNA fragments are detailed in Table S4.

Growth conditions

Cells for strain construction were grown in Luria-Bertani (LB) medium with appropriate antibiotics: ampicillin (100 $\mu\text{g}\cdot\text{ml}^{-1}$), kanamycin (50 $\mu\text{g}\cdot\text{ml}^{-1}$), spectinomycin (50 $\mu\text{g}\cdot\text{ml}^{-1}$), chloramphenicol (25 $\mu\text{g}\cdot\text{ml}^{-1}$), based on plasmid presents. For titratable strains, 10 $\text{ng}\cdot\text{ml}^{-1}$ anhydrotetracycline (aTc) was added during strain construction and seed culture preparation. Seed cultures were initiated by inoculating 3–6 individual colonies from an LB agar plate into 14 ml round-bottom test tubes containing 2 ml of the desired medium for overnight growth.

For physiological phenotypic characterization, all cells were cultivated at 37°C with shaking at 150 rpm in a water bath shaker. The *dnaA*-, *seqA*- and *hns*-titratable cells were characterized under varying aTc concentrations in rich defined medium supplemented with glycerol (M6). *hns*-titratable cells were also characterized in rich defined medium supplemented with glucose (M1). When strains contained reporter plasmid, 25 $\mu\text{g}\cdot\text{ml}^{-1}$ chloramphenicol was added. The autoregulated-*dnaA* strain (MGCL1) with a reporter plasmid, was tested in seven different media: M1 (MOPS + EZ + AUCG + Glucose), M2 (MOPS + EZ + Glucose), M3 (MOPS + EZ + AUCG + Gluconate), M6 (MOPS + EZ + AUCG + Glycerol), M12 (MOPS + EZ + Glycerol), M13 (MOPS + CAA + Glycerol), and M18 (MOPS + Glucose). Detailed medium compositions were provided previously (Zheng et al., 2020).

Steady-state growth was established as previously described (Zheng et al., 2020) by culturing cells for ~10 generations in exponential growth phase with ~15-fold dilution at OD₆₀₀ ~0.2, using the same medium supplemented with antibiotic or aTc as required. OD₆₀₀ measurements were taken at consecutive time points using a spectrometer (Genesys 10s, Thermo Fisher Scientific), and the growth rate (or doubling time) λ was calculated by fitting the data to an exponential growth curve. Samples for determination of population-averaged cell mass ($\bar{m}$) and cellular origin number ($\bar{o}$), quantification of mRNA levels, and mRNA fluorescence in situ hybridization (FISH) were collected when OD₆₀₀ reached ~0.2.

Measurements of population-averaged cellular mass ($\bar{m}$), cellular *oriC* number ($\bar{o}$), the initiation to division period (C+D), and initiation mass (m_i)

The population-averaged cell mass ($\bar{m}$), cellular *oriC* number ($\bar{o}$), and initiation mass (m_i) were calculated as described previously (Zheng et al., 2020 [DOI](#)). Briefly, $\bar{m}$ was determined by dividing the OD₆₀₀ by the cell concentration (OD₆₀₀ ml per 10⁹ cells). For cell counting, samples were collected in precooled cell count buffer (0.9% NaCl with 0.12% formaldehyde, filtered with a 0.22 µm filter) and stained with 1 µg·ml⁻¹ DAPI in the same buffer. Cell number concentration was measured by counting the DAPI-stained particles using flow cytometry equipped with 405 nm laser (CytoFLEX, Beckman Coulter Life Sciences). The $\bar{o}$ was calculated based on the classical run-out experiments (Zheng et al., 2020 [DOI](#)). In brief, cell suspensions were treated with 500 µg·ml⁻¹ rifampicin and 30 µg·ml⁻¹ cephalixin to inhibit new initiation and cell division, respectively, and maintained for 2-3 mass doublings to complete ongoing replication. Cells were fixed with 70% precooled absolute ethanol, and treated with stain buffer (20mM Tris-HCl, pH 8.0, 130 mM NaCl, 10 ng·ml⁻¹ DAPI). Using the same flow cytometry, we measured the cellular origin distribution based on DAPI signaling, from which $\bar{o}$ was calculated. C+D was calculated based on the equation: $C+D = \ln(\bar{o})/\lambda$. m_i was calculated based on the equation: $m_i = \frac{\bar{m}}{\bar{o}} \times \frac{1}{\ln 2}$ (Bremer et al., 1979 [DOI](#); Si et al., 2017 [DOI](#); Zheng et al., 2016 [DOI](#)).

Screening of the synthetic promoter library

After transforming the reporter plasmid into *dnaA*-titratable cells, the seed culture was cultivated in M6 medium with 10 ng·ml⁻¹ aTc, then diluted 1:180 in fresh medium containing either 0.5 or 50 ng·ml⁻¹ aTc and cultivated at 37 °C and 800 r.p.m. in a digital thermo-shaker (AOSHENG) using flat-bottom 96-well plates (Corning). Following approximately 4 hours of pre-cultivation, the cultures were diluted 1:180 in fresh medium with the same aTc concentration. The OD₆₀₀ and GFP fluorescence intensity were continuously monitored during cell growth using a Synergy H1 microplate reader (BioTek). The non-fluorescent strain (MG1655) cultured in 180 µl of M6 medium for each experiment was used to subtract the background fluorescence. For OD₆₀₀ between 0.1 and 0.2, the repression fold change was calculated by dividing the mean fluorescence intensity per OD₆₀₀ of the cells grown with 0.5 ng·ml⁻¹ aTc induction by the corresponding intensity for cells grown with 50 ng·ml⁻¹ aTc induction.

Quantitative real-time PCR

Quantitative real-time PCR was performed using a protocol similar to previously described (Zheng et al., 2016 [DOI](#)). Briefly, RNA was stabilized using Bacteria Protect Reagent (Qiagen) and extracted using RNeasy mini kit (Qiagen). 500 ng RNA was reverse transcribed (TAKARA), and the resulting cDNA samples were diluted 1:25 with PCR-grade water. Quantitative real-time PCR was carried out on a qTower3 (Analytikjena) or CFX Connect system (Bio-Rad), using the following program: 30 s at 95 °C, followed by 40 cycles of denaturation (5 s at 95 °C), annealing, and elongation (30 s at 58 °C). Primer sequences were listed in Table S4. The *rpoA* gene was used as the reference gene.

lacZ and *dnaA* mRNA fluorescence in situ hybridization (FISH)

FISH for *lacZ* or *dnaA* mRNA was performed using a method similar to previously described (Skinner et al., 2013 [DOI](#)). Briefly, antisense DNA oligo probes were designed against *lacZ* or *dnaA*, and synthesized with TAMRA labeling at the 3' end. The sequences of probes targeting *lacZ* or *dnaA* are either as previously described (Skinner et al., 2013 [DOI](#)) or listed in Table S5. 27 ml steady-state growth culture (OD₆₀₀ ~0.2) was fixed by direct adding 3 ml of 37 % (vol/vol) formaldehyde,

following by incubation for 30 min at room temperature with gentle mixing using a nutator. The fixed cells were collected by centrifugation at 600 g for 8 minutes, and washed twice with 1 ml RNase-free PBS. The cell pellet was resuspended in 300 μ l DEPC water, and 700 μ l precooled absolute ethanol was added to permeabilize the cells for 1 hour at room temperature in the nutator. After permeabilization, the cells were divided into two equal parts for characterization of *lacZ* or *dnaA* mRNA as needed. Hybridization was performed as described previously (Skinner et al., 2013), with the stringency of the hybridization and wash solutions adjusted by changing the concentration of formamide, 30 % for *dnaA* probes and 40 % for *lacZ* probes. After hybridization, the cells were washed with the appropriate wash solution and prepared for imaging.

Cell imaging and cell size parameters measurement

For imaging, 3 μ l of cell suspension was pipetted onto the center of a 24 $\times$ 60-mm coverslip. A thin 1.5 % agarose pad (prepared with 1 $\times$ PBS) was then placed on top of the cell suspension droplet. Fluorescent and phase images were acquired using an inverted microscope (IX-83, Olympus) equipped with a 100 $\times$ oil objective (Olympus), an automated xy-stage (ASI, MS2000), and a sCMOS camera (Prime BSI, photometrics). Fluorescence excitation was performed as previously described (Xia et al., 2022). Solid-state lasers (Coherent OBIS: 405–100 LX, 561–100 LS) and Semrock emission bandpass filters (445/40 nm for DAPI and 598/25 nm for RFP) were used for excitation and collecting fluorescent signals.

Images were aligned with phase contrast channel by selecting at least six pairs of common control points and applying a geometric transformation using the MATLAB `imwarp` function. This allowed for adjustment of the cell masks in the RFP or DAPI channel. A customized ImageJ-based image-processing package, MicrobeJ (v5.13l₁) plugin for Fiji (Ducret et al., 2016), was used to contour cells and calculate cell size parameters and fluorescence intensity, including the mean cell width, cell length, and cell area from phase-contrast images. The cell volume was calculated using a cylinder-plus-two-hemispheres geometry. The total RFP intensity per cell volume, corrected for background subtraction, was then used to estimate the mRNA concentration of *lacZ* or *dnaA*.

Characterization of cell cycle-dependent fluctuations in DnaA activity

Data filtering was first performed based on the intensity of the DAPI fluorescence signal. To eliminate non-cell regions resulting from incorrect segmentation of the phase contrast images, only DAPI-positive cells were included in the analysis. To further minimize the potential impact of variability in fixation, permeabilization, and wash efficiencies, which could affect the TAMRA fluorescence signal (representing the mRNA FISH signal), only cells with mean DAPI signals within the 95% confidence intervals were included in the subsequent analysis.

The concentration of the *lacZ* mRNA ($[mZ]$) can be expressed as $[mZ] = J/\delta_Z$, where δ_Z is the degradation rate and J is the mRNA synthesis flux. J is in turn given by the promoter on-rate (k), the RNAP concentration ($[RNAP]$), and the concentration of the *lacZ* gene ($[G_Z]$). Assuming that the factors δ_Z , $[RNAP]$, and $[G_Z]$ are the same between the two strains with P_{syn66} and P_{con} driving *lacZ*, then the ratio of the mRNA concentration obtained at each point of the cell cycle gives the ratio of the promoter on-rates, $k_{P_{syn66}}$ and $k_{P_{con}}$. Since a constitutive promoter is non-responsive to DnaA activity, k_{con} is independent of cell cycle and we can obtain DnaA activity regulated promoter activity as $k_{syn66} = \frac{[mZ](P_{syn66})}{[mZ](P_{con})}$, then the DnaA activity is denoted as k_{syn66}^{-1} .

The concentration of the *lacZ* mRNA were binned based on cell volume, with a bin size of 0.1 μ m³, was used to determine $[mZ]_{(P_{syn66})}$ and $[mZ]_{(P_{con})}$. The relation between $[mZ]_{(P_{con})}$ and cell volume was smoothed using the MATLAB `smooth` function, and from this, k_{syn66} was deduced.

Based on the DnaA activity oscillations, the cell volume where DnaA activity reaches maximum (V^*) was calculated during a representative birth-to-division cell cycle. The cell volume distribution was used to calculate the number of cells in each volume-doubling range. The range containing the highest cells was considered the representative birth-to-division cell cycle and was shaded in gray (Fig. S2 D). From this, V^* was obtained.

Calculation of cell volume at the time of replication initiation (V_i)

Cell volume at the time of replication initiation (V_i) was calculated based on population-averaged cell volume ($\bar{V}$), and the population-averaged cellular *oriC* number ($\bar{o}$). The equation used for this calculation is as follows:

$$V_i = v_i \times N_{oriC}^{ini} = \frac{\bar{V}}{\bar{o}} \times \frac{1}{\ln 2} \times 2^{\lfloor \log_2 \bar{o} \rfloor}$$

where v_i is the initiation volume by definition, i.e. cell volume per *oriC* at the time of replication initiation (Bremer et al., 1979; Si et al., 2017; Zheng et al., 2016). $\bar{V}$ was obtained by analyzing the phase contrast images, $\bar{o}$ was obtained by run-out experiments (Zheng et al., 2020). N_{oriC}^{ini} is the cellular *oriC* number at the time of replication initiation. $\lfloor \log_2 \bar{o} \rfloor$ denotes the floor function of $\log_2 \bar{o}$.

DnaA shut-down experiment

To shut-down DnaA expression, steady-state growth of the CRidnaA1 cells was first established after ~10 generations of exponential growth in M1 medium. At OD₆₀₀ between 0.15 and 0.2, 24 ml of cell culture was transferred to 12 ml of pre-warmed M1 medium containing 150 ng·ml⁻¹ aTc to induce dUn1Cas12f1 expression at time 0. Under the guidance of *sgRNA_{dnaA}*, the transcription of *dnaA* gene was inhibited. Samples were collected at various time points to determine population-averaged cellular mass ($\bar{m}$), cellular *oriC* number ($\bar{o}$), and to quantify mRNA levels. When the OD₆₀₀ reached around 0.2, the cultures were diluted about 2-fold with pre-warmed M1 medium containing 50 ng·ml⁻¹ aTc to maintain adequate nutrition. The steady-state growing MG1655 cells in M1 medium were also collected for quantification of wild-type *dnaA* mRNA levels. Three independent experiments were conducted.

H-NS shift-up experiment

The DnaA activity reporter plasmid P_{syn66}-P_{con}-FPs was transformed into the *hns*-titratable strain (Rhns2), and cells were cultivated overnight in M6 medium containing 10 ng·ml⁻¹ aTc and 25 µg·ml⁻¹ chloramphenicol for seed culture. Seed cultures were washed and grown in the same medium without aTc for ~10 generations of exponential growth. At an OD₆₀₀ of ~0.2, 40 ml of cell culture was transferred to 20 ml of pre-warmed same medium containing 150 ng·ml⁻¹ aTc to induce H-NS expression at time 0. Samples were collected at time points -2, 1, 3, 5, 7, 10, 15, 20, 25, and 30 minutes for the determination of population-averaged cellular mass ($\bar{m}$), cellular *oriC* number ($\bar{o}$), and quantification of mRNA levels. Steady-state growing MG1655 in M6 medium were also collected for *hns* mRNA quantification. Three independent experiments were conducted.

Electrophoretic mobility shift assay (EMSA)

The genes encoding *dnaA* and *hns* from the MG1655 genome were individually amplified and cloned into the expression vector pET-28a using primers listed in Table S4. *E. coli* BL21 (DE3) carrying pET-28a-DnaA or pET-28a-H-NS was used for overproduction of His₆-DnaA or His₆-H-NS. For each protein purification, a 1 L culture was induced by the addition of 1 mM IPTG at OD₆₀₀ = 0.4. After 1.5 hours of incubation, cells were harvested by centrifugation (5,000 rpm for 20 min at 4 °C). The pellet was resuspended in 50 ml of buffer A (50 mM Tris-HCl, pH 7.0, 500 mM NaCl, 0.5 mM PMSF, 5 % glycerol). Cell resuspension was lysed on ice for 30 minutes (3 seconds on, 7 seconds

off) using a sonicator at 4 °C, and cell debris was removed by centrifugation (14,000 rpm for 60 min at 4 °C). Subsequently, the protein was purified from the soluble supernatant by Ni²⁺-affinity chromatography (HisTrap™ HP Columns; GE Healthcare) with a ÄKTA Pure Protein Purification System. Proteins were collected using gradient elution with buffer B [20 mM Tris-HCl, pH 7.5, 500 mM NaCl, 400 mM imidazole, 5 % glycerol]. Eluted proteins were further purified and concentrated by ultrafiltration using Amicon Ultra 10K (for H-NS) and 30K (for DnaA) filters (Millipore). Finally, after overnight dialysis (0.025 µm Membrane Filter; MF-Millipore) at 4 °C, the protein was stored in buffer C [45 mM Hepes-KOH, pH 7.6, 600 mM potassium glutamate, 1 mM DTT, 10 mM magnesium acetate, 0.5 mM EDTA, 20 % glycerol]. Judgement by SDS/PAGE with Coomassie Brilliant blue staining, the purity of each protein was >90 %. For the experiments where H-NS acts on the *datA* bound by DnaA, the purified DnaA, *datA* and ATP were first incubated in buffer D (20 mM Tris-HCl, pH 7.5, 150 mM potassium glutamate, 10 mM magnesium acetate, 8 mM DTT) at 20 °C for 10 min. Then the H-NS was added to buffer D (15 µl) which contained DnaA-*datA* complexes and incubated for additional 10 min at 20 °C. The final complexes were separated by electrophoresis in 2 % (wt/vol) agarose gel at 80 V for 70 min in Tris-Borate buffer, followed by staining with SYBR Safe (Invitrogen).

Titration model

In the titration model, we posit that the cell mass grows exponentially according to the differential equation $\frac{dV}{dt} = \lambda V$, where V represents the volume of the cell and $\lambda = 1.54 \text{ h}^{-1}$ (corresponding to doubling time of 27 min) is the growth rate. Concurrently, the number of total DnaA protein (A) increases at a rate proportional to the cell's mass, described by $\frac{dA}{dt} = \alpha_A V$, with $\alpha_A = 300 \text{ h}^{-1} \mu\text{m}^{-3}$, with $\alpha_A = 300 \text{ h}^{-1} \mu\text{m}^{-3}$ denoting the synthesis rate per unit volume. The chromosome's structure is abstracted as a vector representing the copy numbers of uniformly distributed sites along its length, and the number of DnaA-boxes (A_b) is quantified by summing the boxes number at their respective locations. In this model, 300 DnaA-boxes are evenly distributed on the chromosome while an additional 100 DnaA-boxes are located near the *oriC*. This arrangement approximates a head-weighted distribution of DnaA-boxes and helps prevent over-initiation (Hansen et al., 1991). Binding of DnaA proteins to these DnaA-boxes allows us to calculate the concentration of free DnaA as $[A_f] = (A - A_b)/V$, under condition of $(A - A_b) \geq 0$. DNA replication commences once the concentration of free DnaA exceeds a critical threshold $[A_f^c] = 10 \mu\text{m}^{-3}$: $[A_f] > [A_f^c]$. 10 min sequestration time was set after replication initiation to mimic the sequestration time (Campbell & Kleckner, 1990).

To explore the dynamics of the *oriC* number following the cessation of *dnaA* expression, we can set the synthesis rate to be zero $\alpha_A = 0$. To simulate the dependence of initiation mass on *dnaA* expression, α_A varies from $60 \text{ h}^{-1} \mu\text{m}^{-3}$ to $3840 \text{ h}^{-1} \mu\text{m}^{-3}$.

Extrusion model

The extrusion model was formulated following the titration model, with an additional extruder introduced as a protein with synthesis rate $\alpha_H = 180 \text{ h}^{-1} \mu\text{m}^{-3}$ and dynamics $dH/dt = \alpha_H V$. The additional extruder competes the DnaA-boxes with DnaA proteins, with higher affinity to DnaA-boxes. So that it can displace DnaA from the DnaA-boxes, leading to a revised calculation for free DnaA concentration as $[A_f] = (A - (A_b - H))/V$, under condition of $(A_b - H) \geq 0$ and $(A - (A_b - H)) \geq 0$. To get stochastic modelling for adder phenomenon, we introduced white noises on cell mass growth rate and the DnaA synthesis rate at each replication cycle: $\lambda = \lambda_0 + \eta\lambda_0$, $\alpha_A = \alpha_0 + \eta\alpha_0$ where $\lambda_0 = 0.66 \text{ h}^{-1}$, $\alpha_0 = 300 \text{ h}^{-1} \mu\text{m}^{-3}$ and $\eta = 0.1$ representing white noise. This extrusion model exhibits consistent performance under various parameter ranges.

DnaA-ATP/DnaA-ADP switch model

The switch model was simulated following the idea of LDDR model proposed by Berger *et al.* (Berger & Wolde, 2022). The formulations of the DnaA-ATP fraction dynamics is:

$$\frac{df}{dt} = \left(\frac{\alpha_l}{[D]_T} V + \frac{\alpha_{d1}}{[D]_T} N_{d1} + \frac{\alpha_{d2}}{[D]_T} N_{d2} \right) \frac{1-f}{\frac{K_D}{[D]_T} + 1-f} - \left(\frac{\beta_{datA}}{[D]_T} + \frac{\beta_{rida}}{[D]_T} \right) N_o \frac{f}{\frac{K_D}{[D]_T} + f} + \frac{\alpha_A}{[D]_T} \lambda(1-f)$$

In this equation $[D]_T = 400 \mu\text{m}^{-3}$ is the concentration of total DnaA proteins in a cell $\alpha_l = 750 \text{h}^{-1}$ is the activation rate of DnaA-ATP by lipid; $\alpha_{d1} = 100 \text{h}^{-1}$, $\alpha_{d2} = 643 \text{h}^{-1}$ are the activation rate of DnaA-ATP by the DARS1 and DARS2 sites; $\beta_{datA} = 600 \text{h}^{-1}$, $\beta_{rida} = 500 \text{h}^{-1}$ are the deactivation rate of DnaA-ATP by *datA* and *RIDA*; $K_D = 50 \mu\text{m}^{-3}$ the dissociation constant of DnaA activation and deactivation; N_{d1} , N_{d2} , N_o are copy number of DARS1, DARS2 and *oriC* sites extracted dynamically from the DNA copy number vector; $\alpha_A = 300 \text{h}^{-1} \mu\text{m}^{-3}$ is the DnaA expression level. The DNA replication initiation was determined when f reaches a threshold $f_C = 0.75$ (Berger & Wolde, 2022). To simulate the *dnaA* shut-down, we simply deleted the term of *dnaA* synthesis ($\lambda(1-f)$).

Titration-switch model and titration-switch-extrusion model

In the integrated model combining titration and switch mechanisms, replication initiation is triggered when the concentration of free DnaA-ATP $[A_{ATP}^f]$ exceeds a threshold $[A_{ATP}^{f_c}]$. The concentration of free DnaA-ATP is titrated by the DnaA-boxes on the chromosome as in the titration model, with $[A_{ATP}^f] = (A_{ATP} - A_b)/V$, under condition of $(A_{ATP} - A_b) \geq 0$. The parameters related to the titration dynamics remain consistent with those described in earlier sections of titration model.

The dynamics of total DnaA-ATP (A_{ATP}) is governed by both expression and conversion processes as described in Berger *et al.* (Berger & Wolde, 2022). According to this framework, the lipid and DARS systems facilitate the conversion of DnaA-ADP to DnaA-ATP at constant rates, while the *datA* site and *RIDA* systems also play a role in converting DnaA-ATP to DnaA-ADP, again at fixed rates. Additionally, the synthesis of new DnaA proteins provides another source of DnaA-ATP.

To be specific, DNA replication initiates was set when the concentration of free DnaA-ATP, $[A_{ATP}^f] = (A_{ATP} - A_b)/V$ exceeds a threshold $[A_{ATP}^f]$. The dynamics of total DnaA number in a cell A followed $\frac{dA}{dt} = \frac{\phi_{p0}\lambda V}{1 + ([A_{ATP}^f]/K_D^n)}$, and the dynamics of total DnaA-ATP number A^{ATP} followed:

$$\frac{dA^{ATP}}{dt} = (\alpha_l V + \alpha_{d1} N_{d1} + \alpha_{d2} N_{d2}) \frac{[A^{ADP}]}{K_D + [A^{ADP}]} - (\beta_{datA} + \beta_{rida}) N_o \frac{[A_{ATP}^f]}{K_D + [A_{ATP}^f]} + \frac{\phi_{p0}\lambda V}{1 + ([A_{ATP}^f]/K_D^n)}$$

where the three terms on the right represent activation and deactivation of DnaA-ATP and new synthesis of DnaA-ATP. In these equations $[A^{ADP}] = \frac{(A - A^{ATP})}{V}$ is the concentration of DnaA-ADP; $[D]_T = 400 \mu\text{m}^{-3}$ is the total DnaA concentration; $K_D^p = 400 \mu\text{m}^{-3}$ is the dissociation constant of DnaA promoter; ϕ_{p0} is the DnaA protein ratio relative to all proteins, which is proportional to α_A and ranges from 0.002% to 0.1%; $n = 5$ is the cooperativity of *dnaA* expression; The same activation/deactivation rates and dissociation constants of DnaA (α_l , α_{d1} , α_{d2} , β_{datA} , β_{rida} , K_D) are used as in the switch model; N_{d1} , N_{d2} , N_o are copy number of DARS1, DARS2 and *oriC* sites extracted dynamically from the DNA copy number vector. To get a good fit of the initiation mass-*dnaA* expression relation, high threshold of free DnaA-ATP concentration was applied, $[A_{ATP}^{f_c}] = 200 \mu\text{m}^{-3}$.

To simulate the titration-switch-extrusion model, we incorporated the extruder as described in the extrusion model, where its amount was given by $H = \alpha_H V$. The concentration of free DnaA-ATP was then calculated as $[A_{ATP}^f] = (A_{ATP} - (A_b - H))/V$, under condition of $(A_b - H) \geq 0$ and $(A_{ATP} - (A_b - H)) \geq 0$. To explore the best fit of continuous DNA replication rounds following *dnaA* shut-down, $\alpha_H = 550 \text{ h}^{-1} \mu\text{m}^{-3}$ is used.

All parameters are summarized in Tables S6.

Supplementary information

Note S1

To understand the initiation mass dependency on *dnaA* expression level α_A (**Fig. 2G** [↗](#)), we can solve the dynamics of DnaA amounts $A(t)$ and cell volume $V(t)$ yielding the solutions:

$A(t) = \frac{\alpha_A V_0}{\lambda} e^{\lambda t} + \text{const}$. The boundary conditions corresponding to successive rounds of DNA replication initiation at times $t = 0$ and $t = t_I$ are: $A(t=0) = \frac{A_I \cdot O}{2} + \frac{A_b}{2}$, $V(t=0) = \frac{V_I \cdot O}{2}$, and $A(t=t_I) = A_I \cdot O + A_b$, $V(t=t_I) = V_I \cdot O$, where A_I is the threshold of DnaA number per *oriC* and V_I is the cell volume per *oriC* at the time of initiation (t_I).

Applying these conditions, we derive

$$V_I = \frac{A_I - A_b}{\alpha_A} \lambda$$

showing that the initiation mass is inversely proportional to the DnaA expression level α_A , assuming A_I and A_b are constant. This inverse relationship predicted by the titration model implies a much sharper dependence of initiation mass on DnaA expression than the exponential dependence observed experimentally.

Following the same analysis in the titration model, the extrusion model has the same DnaA amounts and volume dynamics as in the titration model with similar H-NS dynamics

$H(t) = \frac{\alpha_H V_0}{\lambda} e^{\lambda t} + \text{const}$. The condition of DNA replication initiation sets the boundary

condition: $A(t=0) = \frac{A_I \cdot O}{2} + \frac{A_b}{2} + H(t=0)$, $V(t=0) = \frac{V_I \cdot O}{2}$ and

$A(t=t_I) = A_I \cdot O + A_b + H(t=t_I)$, $V(t=t_I) = V_I \cdot O$.

Solving these equations we get:

$$V_I = \frac{A_I - A_b}{\alpha_A + \alpha_H} \lambda$$

The adding of extruder shifts the inverse dependency of initiation mass - dnaA expression curve to the left which results in a flatter curve. To get a better fitting to experiment data, $\alpha_H = 1500 \text{ h}^{-1} \mu\text{m}^{-3}$ were used in **Fig. 6G** [↗](#).

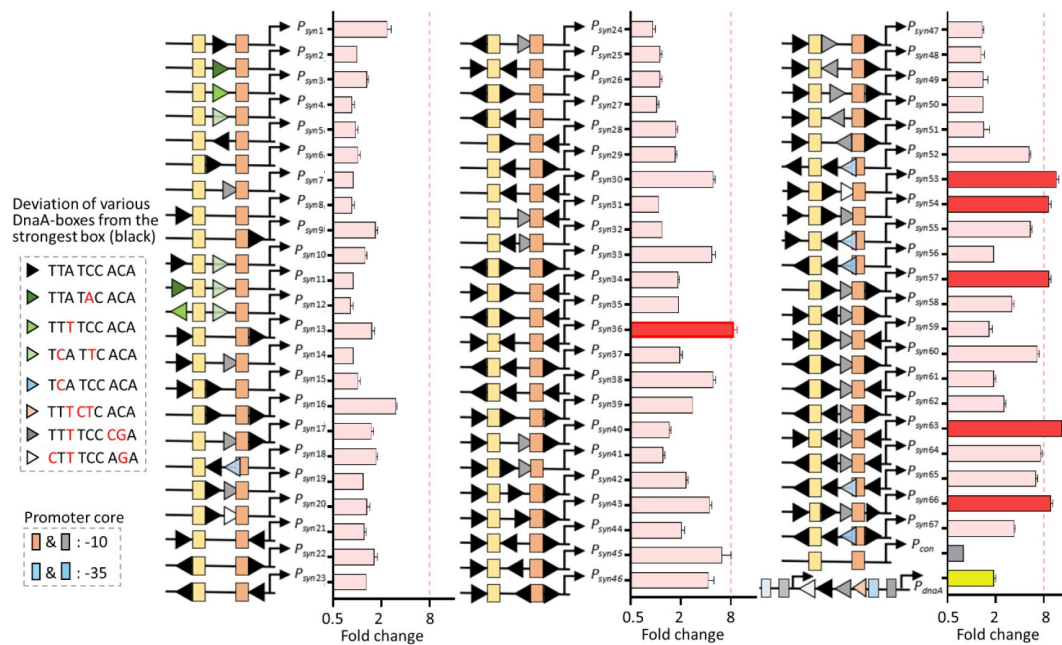


Figure S1.

Screening a library of synthetic promoters reveals potential candidates to report DnaA activity.

For each synthetic promoter, various combinations of DnaA-boxes were inserted near the promoter core, and their repression fold in response to elevated DnaA expression was measured in *dnaA*-titratable cells grown in 0.5 ng·ml⁻¹ aTc compared to cells grown in 50 ng·ml⁻¹ aTc, using a microplate reader. Error bars represent the standard deviation (SD) of three replicates.

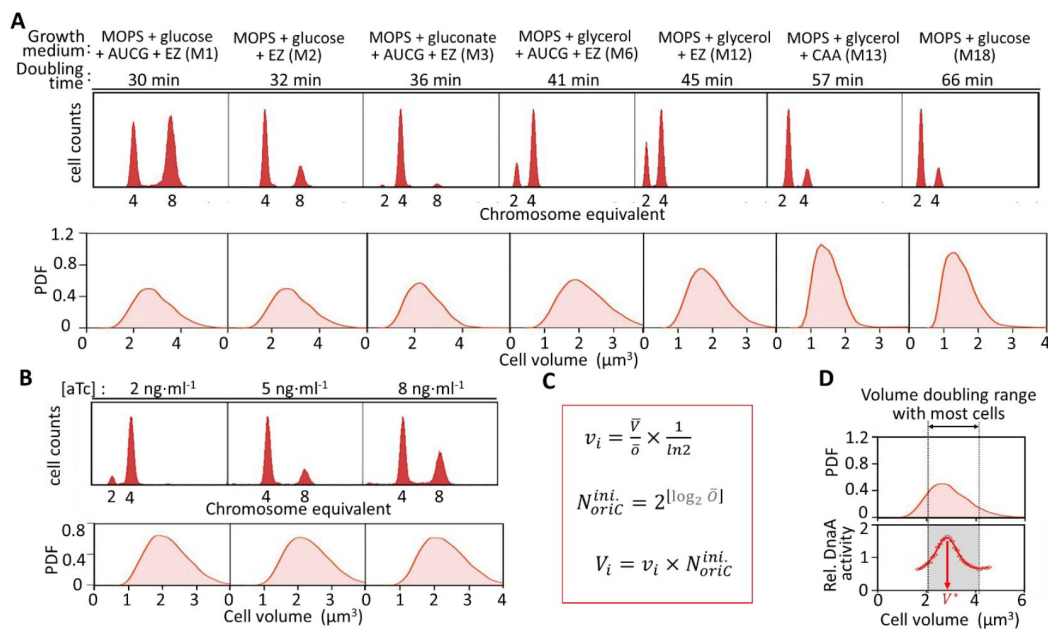


Figure S2.

Determination of cell volume at replication initiation and representative birth-to-division cell cycle.

(A-B) Distribution of cellular *oriC* content (up) and cell volume (low) for cells shown in Fig.4C cultivated in various growth media with noted doubling time **(A)**, and for cells shown in Fig. 4F grown in M6 medium with noted aTc concentrations **(B)**. Cellular *oriC* was determined using run-out experiments followed by flow cytometry, and cell volume was obtained from phase-contrast images from mRNA FISH experiments. **(C)** Calculation of the cell volume at replication initiation. Based on panels **A** and **B**, the population-averaged cellular *oriC* content ($\bar{\delta}$) and cell volume ($\bar{v}$) were determined, then the initiation volume can be deduced accordingly. Additionally, the *oriC* number at the initiation time (N_{oriC}^{ini}) was calculated based on $\bar{\delta}$. Cell volume at the replication initiation was derived by definition. **(D)** Illustration of the determination of the representative birth-to-division cell cycle for DnaA activity oscillations. The cell volume distribution is analyzed to calculate the number of cells within each volume-doubling range. The range with the most cell counts is defined as the representative birth-to-division cycle, which is shaded in gray.

Figure S3.

Predictions of the extrusion model.

(A) Dynamics of relative total cell mass and *oriC* number (upper panel), *oriC* concentration (middle panel), and free DnaA concentration (lower panel), predicted by a deterministic version of the extrusion model, without considering cell division. **(B)** Simulations of DnaA shut-down. DnaA shut-down was simulated by changing the *dnaA* expression level α_A to 0 (upper panel). During DnaA shut down, dynamics of relative *oriC* concentration (middle panel) and free DnaA concentration (lower panel) predicted by the titration model and extrusion model were plotted and compared. **(C)** Initiation adder phenomenon predicted by the extrusion model. A stochastic version of the extrusion model was simulated. The next initiation mass (upper panel), added mass between successive initiations (middle panel), and added time between successive initiations (lower panel) were plotted against the initiation mass in previous initiation, both normalized to their averages (blue dots, $n=820$). Mean values were calculated in bins of 0.05 relative initiation mass (red dotted line)

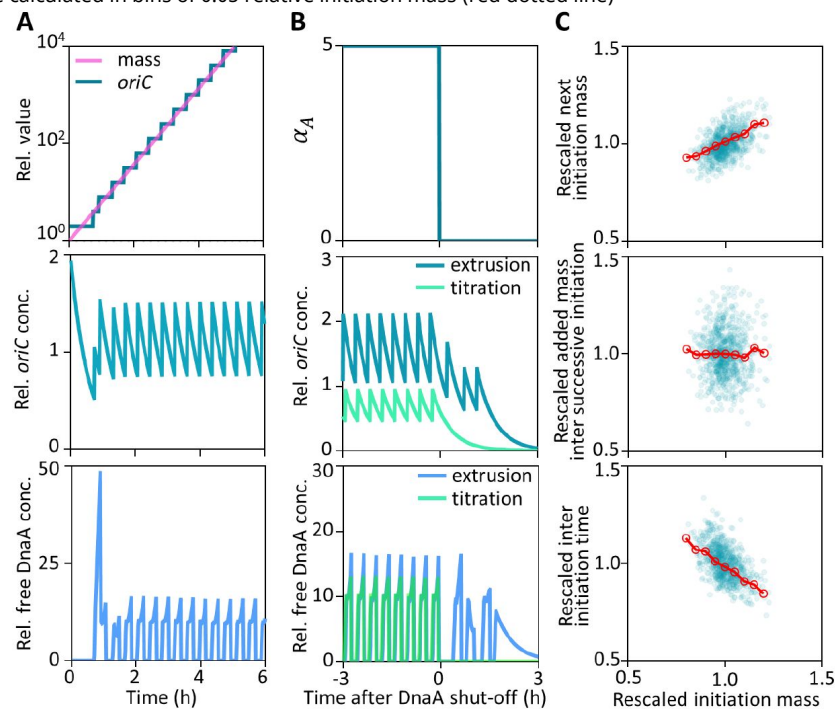


Figure S4.

H-NS promotes the release of DnaA from the *datA* sequence.

Competitive binding between H-NS and DnaA to *datA* fragments. H-NS and DnaA formed distinct complexes with *datA*, resulting in H-NS:*datA* (lane 2) and DnaA:*datA* (lane 3) complexes. Addition of H-NS to pre-incubated DnaA:*datA* complex resulted in coexistence of both DnaA:*datA* and H-NS:*datA* complexes (lane 4).

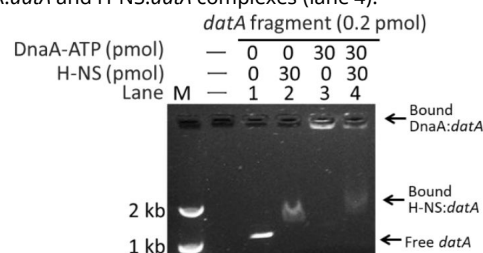


Figure S5.

Basic phenotypic characterization of *hns*-titratable cells harboring a DnaA activity reporting plasmid.

(A) Relative *hns* mRNA levels were measured in steady-state growing cells in rich defined medium supplemented with glycerol at various aTc concentration. Growth rate (B), population-averaged cellular mass $\bar{m}$ (C), and cellular *oriC* number $\bar{o}$ (D) were characterized as a function of *hns* mRNA levels for these steady-state cultures. Relative DnaA activity (represented by free DnaA concentration) (E) and initiation mass (F) as a function of varying extruder concentration as predicted by the extrusion model.

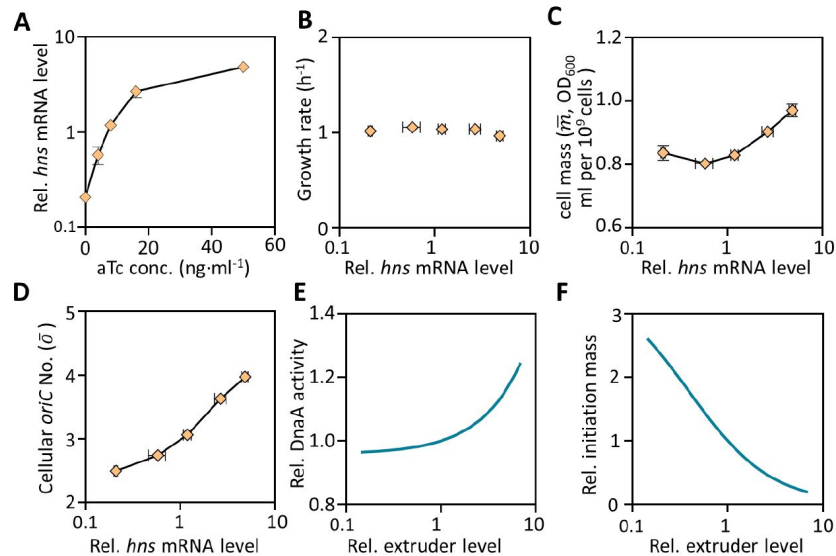


Figure S6.

Predictions of the titration-switch model and titration-switch-extrusion model.

(A) Comparison of the titration-switch model predictions and experimental data for the relation between relative initiation mass and *dnaA* mRNA levels. The relative initiation mass was calculated as mass/*oriC* averaged over more than 100 cycles after steady DNA replication initiation was established, while the relative *dnaA* mRNA was achieved by setting various *dnaA* expression rate α_A . Dynamics of total *oriC* number during DnaA shut-down predicted by titration-switch model (B) and titration-switch-extrusion model (C), normalized to the value at DnaA shut-down (dash line).

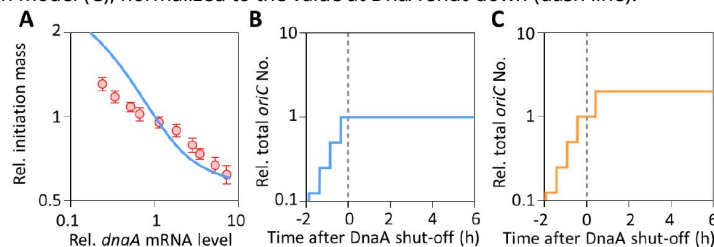


Table S1.

Strains used in this study

Strain	Relevant genetic marker(s) or features	Source or reference
MG1655	<i>E. coli</i> K12(AMB1655)	(Liu et al., 2011)
CL1	MG1655 $\Delta cheZ$, Δlac	(Liu et al., 2011)
MGCL1	MG1655 Δlac	This study
RdnaA1	MG1655 P_{dnaA} - <i>dnaA</i> :: P_{kanR} - <i>kanR</i> <i>yidA</i> ::(<i>bla</i> : P_{tet} - <i>dnaA</i>):: <i>yidX</i> <i>intS</i> :: P_{tet} - <i>tetR</i>	This study
RdnaA2	RdnaA1 Δlac	This study
RseqA1	MG1655 <i>seqA</i> :: <i>kan</i> <i>attB</i> ::(<i>bla</i> : P_{tet} - <i>tetR</i> - <i>seqA</i>)	This study
Rhns1	MG1655 <i>hns</i> :: <i>kan</i> <i>attB</i> ::(<i>bla</i> : P_{tet} - <i>tetR</i> - <i>hns</i>)	This study
Rhns2	Rhns1 Δlac	This study
CRidnaA1	MG1655 <i>asnA</i> ::(P_{J23119} - <i>sgRNA</i> _{<i>dnaA</i>} : P_{J23100} - <i>tetR</i> : P_{tet} - <i>dUn1Cas12f1</i>):: <i>viaA</i>	This study

Table S2.

Plasmids used in this study

Plasmid	Relevant genotype	Source or reference
pSIM5	Cm^r , <i>repA101</i> (<i>Ts</i>) <i>ori</i> , λ Red	(Zheng et al., 2016)
plkml	Amp^r , pUC <i>ori</i> , <i>loxP</i> - <i>kan</i> - <i>loxP</i>	(Zheng et al., 2016)
pMD19-tetR	Amp^r , pUC <i>ori</i> , <i>bla</i> : P_{tet} - <i>tetR</i>	(Zheng et al., 2016)
pMD19-hupA-mcherry	Amp^r , pUC <i>ori</i> , <i>bla</i> : P_{tet} - <i>tetR</i> - <i>hupA</i> - <i>mcherry</i>	This study
pMD19-Rhns	Amp^r , pUC <i>ori</i> , <i>bla</i> : P_{tet} - <i>tetR</i> - <i>hns</i>	This study
pMD19-RseqA	Amp^r , pUC <i>ori</i> , <i>bla</i> : P_{tet} - <i>tetR</i> - <i>seqA</i>	This study
pMD19-RdnaA	Amp^r , pUC <i>ori</i> , <i>bla</i> : P_{tet} - <i>tetR</i> - <i>dnaA</i>	This study
p15A-RdnaA	Kan^r , p15A <i>ori</i> , P_{tet} - <i>tetR</i> - <i>dnaA</i>	This study
pTargetF	<i>aadA'</i> , pMB1 <i>ori</i> , <i>sgRNA</i>	(Jiang et al., 2015)
pEcCas	Kan^r , pSC101 <i>ori</i> , <i>sacB</i> P_{araB} - λ Red P_{cas} - <i>cas9</i>	(Li et al., 2021)
pZA31- P_{tet} -M2-GFP	Cm^r , p15A <i>ori</i> , P_{tet} - <i>gfp</i>	(Liu et al., 2019)
CmPcas	Cm^r , pSC101 <i>ori</i> , <i>sacB</i> P_{araB} - λ Red P_{cas} - <i>cas9</i>	This study
CPP00458	<i>aadA'</i> , pSC101 <i>ori</i> , P_{J23119} - <i>sgRNA</i> - T - P_{J23100} - <i>tetR</i> - T - P_{tet} - <i>dUn1Cas12f1</i>	Gift from Fu lab
P_CRidnaA1	<i>aadA'</i> , pSC101 <i>ori</i> , P_{J23119} - <i>sgRNA</i> _{<i>dnaA</i>} - T - P_{J23100} - <i>tetR</i> - T - P_{tet} - <i>dUn1Cas12f1</i>	This study
pPT	Cm^r , pSC101 <i>ori</i> , <i>Bsal</i> - <i>lacZ</i> α - <i>Basl</i> - <i>riboJ</i> - <i>sfgfp</i>	(Zong et al., 2017)
pPT-RFP	Cm^r , pSC101 <i>ori</i> , <i>Bsal</i> - <i>lacZ</i> α - <i>Basl</i> - <i>riboJ</i> - <i>mcherry</i>	This study
pPT-lacZ	Cm^r , pSC101 <i>ori</i> , <i>Bsal</i> - <i>lacZ</i> α - <i>Basl</i> - <i>riboJ</i> - <i>lacZ</i>	This study
P_{syn66} -GFP	Cm^r , pSC101 <i>ori</i> , P_{syn66} - <i>riboJ</i> - <i>sfgfp</i>	This study
P_{con} -GFP	Cm^r , pSC101 <i>ori</i> , P_{con} - <i>riboJ</i> - <i>sfgfp</i>	This study
P_{syn66} -RFP	Cm^r , pSC101 <i>ori</i> , P_{syn66} - <i>riboJ</i> - <i>mcherry</i>	This study
P_{native} -GFP	Cm^r , pSC101 <i>ori</i> , P_{native} - <i>riboJ</i> - <i>sfgfp</i>	This study
P_{syn66} -lacZ	Cm^r , pSC101 <i>ori</i> , P_{syn66} - <i>riboJ</i> - <i>lacZ</i>	This study
P_{con} -lacZ	Cm^r , pSC101 <i>ori</i> , P_{con} - <i>riboJ</i> - <i>lacZ</i>	This study
P_{syn66} - P_{con} -FPs	Cm^r , pSC101 <i>ori</i> , P_{con} - <i>riboJ</i> - <i>sfgfp</i> , P_{syn66} - <i>riboJ</i> - <i>mcherry</i>	This study
pET-28a-DnaA	Kan^r , pUC <i>ori</i> <i>f1</i> <i>ori</i> , P_{lacI} - <i>lacI</i> , $P_{T7/lacO}$ - <i>dnaA</i> -6* <i>his</i>	This study
pET-28a-H-NS	Kan^r , pUC <i>ori</i> <i>f1</i> <i>ori</i> , P_{lacI} - <i>lacI</i> , $P_{T7/lacO}$ - <i>hns</i> -6* <i>his</i>	This study

Primers	Sequence	Use
DP420	cctggtagatagattgacaagagttatccacagtaggatactgagcaca	P_{syn1}
DP421	agcttgctcagtagtctactgtggataactctgtcaatctatctac	
DP422	cctggtagatagattgacactgtgtatacacagggcgatactgagcaca	P_{syn2}
DP423	agcttgctcagtagtgcctgtgtataacaagtgtaactctatctac	
DP424	cctggtagatagattgacaataactttccacaggtagatactgagcaca	P_{syn3}
DP425	agcttgctcagtagtctactgtggaaaagtattgtcaatctatctac	
DP426	cctggtagatagattgacaccgatcattcacagtagatactgagcaca	P_{syn4}
DP427	agcttgctcagtagtctaactgtgaatgatcggtgtcaatctatctac	
DP428	cctggtagatagattgacactgtgtggataagggcgatactgagcaca	P_{syn5}
DP429	agcttgctcagtagtgccttatccacacaagtgtaactctatctac	
DP430	cctggtagatagattgacattatccacatagttcccgatactgagcaca	P_{syn6}
DP431	agcttgctcagtagtgcggaaactatgtgataatgtcaatctatctac	
DP432	cctggtagatagattgacaccctgcgattttcccgatactgagcaca	P_{syn7}
DP433	agcttgctcagtagtgcggaaaaatcgagggtgtcaatctatctac	
DP434	cctgtatccacattgacaccctgcgtagttcccgatactgagcaca	P_{syn8}
DP435	agcttgctcagtagtgcggaaactatcgagggtgtcaatgtggataa	
DP436	cctggtagatagattgacaccctgcgtagttcccgatactttatccac	P_{syn9}
DP437	agcttggtgataaagtatcggaactatcgagggtgtcaatctatctac	
DP438	cctgacagagttatccacagtagatagattgacaccgatcattcacagtagatactgagcaca	P_{syn10}
DP439	agcttgctcagtagtctaactgtgaatgatcggtgtcaatctatctactgtggataactctgt	
DP440	cctggaggggtatatacaactcaagattgacaccgatcattcacagtagatactgagcaca	P_{syn11}
DP441	agcttgctcagtagtctaactgtgaatgatcggtgtcaatctttgagttgtataaccctc	
DP442	cctgccatactgtggaaaaggtagaagattgacaccgatcattcacagtagatactgagcaca	P_{syn12}
DP443	agcttgctcagtagtctaactgtgaatgatcggtgtcaatcttctacctttccacagtagtg	
DP444	cctgtatccacattgacaccctgcgtagttcccgatactttatccac	P_{syn13}
DP445	agcttggtgataaagtatcggaactatcgagggtgtcaatgtggataa	
DP446	cctgtatccacattgacaccctgcgattttcccgatactgagcaca	P_{syn14}
DP447	agcttgctcagtagtgcggaaaaatcgagggtgtcaatgtggataa	
DP448	cctgtatccacattgacattatccacatagttcccgatactgagcaca	P_{syn15}
DP449	agcttgctcagtagtgcggaaactatgtgataatgtcaatgtggataa	
DP450	cctggtagatagattgacattatccacatagttcccgatactttatccac	P_{syn16}
DP451	agcttggtgataaagtatcggaactatgtggataatgtcaatctatctac	
DP452	cctggtagatagattgacaccctgcgattttcccgatactttatccac	P_{syn17}
DP453	agcttggtgataaagtatcggaaaaaatcgagggtgtcaatctatctac	
DP454	cctggtagatagattgacattgtggataaagttggatgatactgagcaca	P_{syn18}
DP455	agcttgctcagtagtcatccacacattatccacatgtcaatctatc	
DP456	cctggtagattgacattatccacagttttcccgatactgagcaca	P_{syn19}
DP457	agcttgctcagtagtgcggaaaaactgtggataatgtcaatctatc	
DP458	cctggtagattgacattatccacagctttccagatactgagcaca	P_{syn20}
DP459	agcttgctcagtagtctggaagctgtggataatgtcaatctatc	
DP460	cctgtatccacattgacaccctgcgtagttcccgatactgtggataa	P_{syn21}
DP461	agctttatccacagtagtgcggaaactatcgagggtgtcaatgtggataa	
DP462	cctgtgtggataattgacaccctgcgtagttcccgatactttatccac	P_{syn22}
DP463	agcttggtgataaagtatcggaactatcgagggtgtcaattatccaca	
DP464	cctgtgtggataattgacaccctgcgtagttcccgatactgtggataa	P_{syn23}
DP465	agctttatccacagtagtgcggaaactatcgagggtgtcaattatccaca	
DP466	cctgtgtggataattgacaccctgcgattttcccgatactgagcaca	P_{syn24}
DP467	agcttgctcagtagtgcggaaaaatcgagggtgtcaattatccaca	
DP468	cctgtatccacattgacattgtggataaagttcccgatactgagcaca	P_{syn25}
DP469	agcttgctcagtagtgcggaaactattatccacatgtcaatgtggataa	
DP470	cctgtgtggataattgacattatccacatagttcccgatactgagcaca	P_{syn26}
DP471	agcttgctcagtagtgcggaaactatgtggataatgtcaattatccaca	
DP472	cctgtgtggataattgacattgtggataaagttcccgatactgagcaca	P_{syn27}
DP473	agcttgctcagtagtgcggaaactattatccacatgtcaattatccaca	
DP474	cctggtagatagattgacattatccacatagttcccgatactgtggataa	P_{syn28}
DP475	agctttatccacagtagtgcggaaactatgtggataatgtcaatctatctac	

Table S3.

Primer pairs for synthetic reporter

DP476	cctggtagatagattgacatgtggataatagtcccagatactttatccac	P_{syn29}
DP477	agctgtggataaagtatcgggaactattatccacatgtcaatctatctac	
DP478	cctggtagatagattgacatgtggataatagtcccagatactgtggataa	P_{syn30}
DP479	agctttatccacagtatcgggaactattatccacatgtcaatctatctac	
DP480	cctggtagatagattgacacccctgcgattttcccagatactgtggataa	P_{syn31}
DP481	agctttatccacagtatcgggaaaaatcgagggtgtcaatctatctac	
DP482	cctggtagatagattgacatgtggataatttcccagatactgagcaca	P_{syn32}
DP483	agcttgctcagtatcgggaaaaatattccacatgtcaatctatctac	
DP484	cctgtgtggataattgacattatccacatagtcccagatactttatccac	P_{syn33}
DP485	agctgtggataaagtatcgggaactatgtggataatgtcaattatccaca	
DP486	cctgttatccacattgacattatccacatagtcccagatactgtggataa	P_{syn34}
DP487	agctttatccacagtatcgggaactatgtggataatgtcaatgtggataa	
DP488	cctgtgtggataattgacattatccacatagtcccagatactgtggataa	P_{syn35}
DP489	agctttatccacagtatcgggaactatgtggataatgtcaattatccaca	
DP490	cctgttatccacattgacatgtggataatagtcccagatactgtggataa	P_{syn36}
DP491	agctttatccacagtatcgggaactattatccacatgtcaatgtggataa	
DP492	cctgtgtggataattgacatgtggataatagtcccagatactttatccac	P_{syn37}
DP493	agctgtggataaagtatcgggaactattatccacatgtcaattatccaca	
DP494	cctgtgtggataattgacatgtggataatagtcccagatactgtggataa	P_{syn38}
DP495	agctttatccacagtatcgggaactattatccacatgtcaattatccaca	
DP496	cctgttatccacattgacattatccacatagtcccagatactttatccac	P_{syn39}
DP497	agctgtggataaagtatcgggaactatgtggataatgtcaatgtggataa	
DP498	cctgttatccacattgacacccctgcgattttcccagatactttatccac	P_{syn40}
DP499	agctgtggataaagtatcgggaaaaatcgagggtgtcaatgtggataa	
DP500	cctgttatccacattgacattatccacagttttcccagatactgagcaca	P_{syn41}
DP501	agcttgctcagtatcgggaaaaactgtggataatgtcaatgtggataa	
DP502	cctgttatccacattgacacccctttatccacacccgatactttatccac	P_{syn42}
DP503	agctgtggataaagtatcgggtgtggataaagggtgtcaatgtggataa	
DP504	cctgttatccacattgacacccctgcgttatccacagatactttatccac	P_{syn43}
DP505	agctgtggataaagtatcgtggataacgcagggtgtcaatgtggataa	
DP506	cctgttatccacattgacatgtggataatagtcccagatactttatccac	P_{syn44}
DP507	agctgtggataaagtatcgggaactattatccacatgtcaatgtggataa	
DP508	cctgttatccacattgacaccccttggataaaccgatactttatccac	P_{syn45}
DP509	agctgtggataaagtatcgggttatccacaagggtgtcaatgtggataa	
DP510	cctgttatccacattgacacccctgcgtgtggataagatactttatccac	P_{syn46}
DP511	agctgtggataaagtatcttatccacacgcagggtgtcaatgtggataa	
DP512	cctgttatccacattgacattttccgatagttcccagatactttatccac	P_{syn47}
DP513	agctgtggataaagtatcgggaactatcgggaaaatgtcaatgtggataa	
DP514	cctgttatccacattgacatcgggaaaatagtcccagatactttatccac	P_{syn48}
DP515	agctgtggataaagtatcgggaactattttccgagatgtcaatgtggataa	
DP516	cctgttatccacattgacaccccttttcccagaccgatactttatccac	P_{syn49}
DP517	agctgtggataaagtatcgggtcgggaaaaagggtgtcaatgtggataa	
DP518	cctgttatccacattgacaccccttgggaaaaccgatactttatccac	P_{syn50}
DP519	agctgtggataaagtatcgggttttcccgaagggtgtcaatgtggataa	
DP520	cctgttatccacattgacacccctgcgtcgggaaaagatactttatccac	P_{syn51}
DP521	agctgtggataaagtatctttccgacgcagggtgtcaatgtggataa	
DP522	cctgtgtggataaagatagattgacatgtggataagtgtggatgatactgagcaca	P_{syn52}
DP523	agcttgctcagtatcatccacacttatccacatgtcaatctatcttatccaca	
DP524	cctgttatccacagatagattgacattatccacagctttccagatactgagcaca	P_{syn53}
DP525	agcttgctcagtatcggaaaagctgtggataatgtcaatctatctgtggataa	
DP526	cctgttatccacagatagattgacattatccacagttttcccagatactgagcaca	P_{syn54}
DP527	agcttgctcagtatcgggaaaactgtggataatgtcaatctatctgtggataa	
DP528	cctgttatccacagatagattgacatgtggataagtgtggatgatactgagcaca	P_{syn55}
DP529	agcttgctcagtatcatccacacttatccacatgtcaatctatctgtggataa	
DP530	cctgtgtggataattgacatgtggataagtgtggatgatactgagcaca	P_{syn56}
DP531	agcttgctcagtatcatccacacttatccacatgtcaatctatccaca	

Table S3. (continued)

DP532	cctgttgacattatccacagttttccgatactttatccac	
DP533	agctgtggataaaagtatcgggaaaaactgtggataatgtcaa	$P_{\text{syn}57}$
DP534	cctgttatccacattgacatgtggataattttccgatactttatccac	
DP535	agctgtggataaaagtatcgggaaaaattatccacatgtcaatgtggataa	$P_{\text{syn}58}$
DP536	cctgttatccacattgacattatccacatttttccgatactgtggataa	
DP537	agctttatccacagtatcgggaaaaatgtggataatgtcaatgtggataa	$P_{\text{syn}59}$
DP538	cctgtgtggataattgacattatccacatttttccgatactttatccac	
DP539	agctgtggataaaagtatcgggaaaaatgtggataatgtcaattatccaca	$P_{\text{syn}60}$
DP540	cctgtgtggataattgacattatccacatttttccgatactgtggataa	
DP541	agctttatccacagtatcgggaaaaatgtggataatgtcaattatccaca	$P_{\text{syn}61}$
DP542	cctgtgtggataattgacatgtggataattttccgatactttatccac	
DP543	agctgtggataaaagtatcgggaaaaattatccacatgtcaattatccaca	$P_{\text{syn}62}$
DP544	cctgttatccacattgacatgtggataattttccgatactgtggataa	
DP545	agctttatccacagtatcgggaaaaattatccacatgtcaatgtggataa	$P_{\text{syn}63}$
DP546	cctgtgtggataattgacatgtggataattttccgatactgtggataa	
DP547	agctttatccacagtatcgggaaaaattatccacatgtcaattatccaca	$P_{\text{syn}64}$
DP548	cctgtgtggataattgacatgtggataaagtgtggatgatactgtggata	
DP549	agcttatccacaagtatcatccacattatccacatgtcaattatccaca	$P_{\text{syn}65}$
DP550	cctgttatccacattgacattatccacagttttccgatactttatccac	
DP551	agctgtggataaaagtatcgggaaaaactgtggataatgtcaatgtggataa	$P_{\text{syn}66}$
DP552	cctgtgtggataattgacatgtggataaagtgtggatgatactttatccac	
DP553	agctgtggataaaagtatcatccacattatccacatgtcaattatccaca	$P_{\text{syn}67}$
DP554	cctgttgacacccctgcgatagttccgatactgagcaca	
DP555	agcttgtgctcagtatcgggaactatcgaggggtgtcaa	P_{con}
DP556	cctgttatccacacccgggttatccacagttttccgagcccttatccac	
DP557	agctgtggataagggtctcgggaaaaactgtggataacccgggtgtggataa	P_{neg}

Table S3. (continued)

Primers	Sequence	Use
DJP001	gaaagaggagaaatactagatgacccatgattacggttcac	Amplifying <i>lacZ</i> gene from MG1655 genome
DJP002	ttgatgcctgcttattcattttttgacaccagaccaact	
DJP003	gaaagaggagaaatactagatggtttccaaggcgagg	Amplifying mCherry gene from pMD19-hupA-mcherry
DJP004	ttgatgcctgcttattcattttgtagagctcatccatgc	
DJP005	ccacaaggtctccagctgatcaagatcctgcaaaacgat	Amplifying native <i>dnaA</i> promoter (P_{native}) from MG1655 genome
DJP006	ccacatggctccccctgccaattttgtctatggtcat	
DJP007	ctgtttcttgcaagattactagtcacatcagtcattgtacagttcatcatacc	Amplifying $P_{\text{con}}\text{-gfp}$ cassette from $P_{\text{con}}\text{-GFP}$ plasmid
DJP008	ccttagtgactctgagtcctgggtgtgacacccctgcgat	
DJP009	cgccatattgctactttcgtttggca	Amplify <i>dnaA</i> gene from MG1655 genome for the construction of pET-28a-DnaA plasmid
DJP010	cgcaagcttttacgatgacaatgttctga	
DJP011	cgccatattgagcggaagcacttaaaat	Amplify <i>hns</i> gene from MG1655 genome for the construction of pET-28a-H-NS plasmid
DJP012	cgcaagctttattgcttgatcaggaaatc	
DJP013	cgatctgcagaaaggagaaatactaggtgtcactttcgtttggc	Amplifying <i>dnaA</i> gene from MG1655 genome
DJP014	gtgagccggatcctacgatgacaatgttctgatt	
DJP015	atgcctgcagtcacacaggaacactactagatgaaaacgtgaagttgatgatg	Amplifying <i>seqA</i> gene from MG1655 genome
DJP016	gtcaggatccttagatagttcgcgaacacctct	
DJP017	cagtaagcttaagaggagaaatactagatgagcggaagcacttaaaatc	Amplifying <i>hns</i> gene from MG1655 genome
DJP018	gtacggatccttattgcttgatcaggaaatcgctg	
DJP019	tggatcgcaagaagggc	Amplifying T3- $P_{\text{tet}}\text{-tetR-dnaA}$ cassette from pMD19-RdnaA plasmid
DJP020	tcgatatcaaccatggctcgggcaaaatcgctcgagt	
DJP021	cgttttattgatgggtcgacctgcagggaagccacgt	Amplifying <i>kanr</i> gene from the plasmid pEcCas
DJP022	ttcttcgcatcatgctagcagcaaccaattaaccaatc	
DJP023	gcagccatggttgatctgagctcgcttga	Amplifying p15A-T ₃ fragment from the PZA31- P_{tet} -M2-GFP plasmid
DJP024	cacatgaagtcgacccatcaataaaacgaaaggctc	
DJP025	tcaaccactgcagcaaccaattaaccaattctgattacgccccgccttgcca	Amplifying Cm ^r gene from pZA31- P_{tet} -M2-GFP plasmid for the construction of CmPcas
DJP026	tgtctgttacataaacagtaatacaaggggtgttatggagaaaaaacactgg	
DJP027	ttaaaggtattaaaaacaacttttgtctttttacctccgttgcctcaggaaacagctatgacatg	Amplifying the T ₀ -Amp-T ₁ - $P_{\text{tet}}\text{-tetR-hns}$ or T ₀ -Amp-T ₁ - $P_{\text{tet}}\text{-tetR-seqA}$ cassette, and then assembling to <i>attB</i> locus via λ Red recombination system
DJP028	cacaggtgtctcgggctatgaaatagaaaaatgaatcgttgaagcctgtgtaaacgacggccagt	
DJP029	tctattattactcaacaaccaccaataaagtttgagattactacaatgattgaacaa gatggattgcac	Amplifying <i>kanr</i> gene from plkml plasmid for the deletion of native <i>hns</i> gene
DJP030	aaaaaatcccgccgctggcggttttaagcaagtcacaaaagatcagaagaa ctgcaagaagg	
DJP031	ggcctgcacgattggtgattgccattgcttcttctgtctgcaacgttctagtgaacctctt cgaggg	Amplifying <i>kanr</i> gene from plkml plasmid for the deletion of native <i>seqA</i> gene
DJP032	catatactcctgctgactgtattcagctaaacacgactgcactggattaaggccgatcatatt caataacc	
DJP033	gagtcattgcacagattcgta	T ₀ - $P_{\text{tet}}\text{-tetR-T3}$ cassette was amplified from plasmid pMD19-tetR with primers DJP035/DJP036, the <i>intS</i> upstream and downstream sequences were amplified from MG1655 genome with primers DJP033/DJP034, and DJP037/DJP038, respectively. Three fragments were ligated via overlap PCR with primers DJP033/DJP038.
DJP034	gatctgaagcgaaacatga	
DJP035	tcatggttcgcttcagatcaggtgtgttctctctcattc	
DJP036	cctcatagccgattgtttgaaggaaacagctatgacctga	
DJP037	caaacaatcggtatgagg	
DJP038	agtgtataagggtgttcagc	
DJP039	gtacgttagatcgtagacgtttggcgataaagaacgccattcggcgccgtgagcattt aggatccggctcacttca	Amplifying T3-kanr-T1 cassette from p15A-RdnaA plasmid to replace the native $P_{\text{native}}\text{-dnaA}$ on the genome
DJP040	gatcgattaagccaattttgtctatggtcattaaatttccaatagcgcgtaaatctagg gcgcggtatttg	
DJP041	ttaggcacccagccttac	The <i>dnaA-T3</i> and T ₀ -Amp-T ₁ - P_{tet} fragments were amplified from the pMD19-RdnaA plasmid using primers DJP041/DJP042 and DJP043/DJP044, respectively. These two fragments were then combined via overlap PCR using primers DJP045/DJP046 to generate a homologous recombination fragment for inserting T ₀ -Amp-T ₁ - $P_{\text{tet}}\text{-dnaA-T3}$ between the <i>uidA</i> and <i>uidX</i> genes.
DJP042	cagtgatagatactgagcacataagcttaagaggagaaagactaggtgtcactttcgc	
DJP043	gtgctcagtatctctactgatagggatgcaatctctatcactgataggaggactcgag	
DJP044	aaaacgacggcagtgaa	
DJP045	gatggcgtggcgtttgctattgagaagtgtgctgaattaatctgtggcggtcatcttcggtactgtct	Inserting to the CPP00458 backbone to generate P_CridnaA1 plasmid to shut down <i>dnaA</i> expression
DJP046	accgctgcaatttctgtgtatgagtaaaaccaataatcagtaagcgaggaacagctatgacctg	
DJP047	saacttcgagtgaggtccgctg	
DJP048	cgagcacggcggaactccactcgaa	

Table S4.

Oligonucleotides for the construction of strains and plasmids

Table S4. (continued)

DJP049	aaacgatgaagaccgtctttctcc	Amplifying homologous sequences to <i>asnA</i> gene
DJP050	ttcttagacgtcaggtggcattattacagcagagaaggagcg	
DJP051	tgccacctgacgtctaagaa	Amplifying <i>P</i> _{123119-SgRNA_{dnaA}:<i>P</i>_{123100-tetR:<i>P</i>_{tet-dUn1Cas12f1} cassette from P_CRidnaA1 plasmid}}
DJP052	tctagattactgcgcagatggcgacgataatgacagcagccaactcagcttc	
DJP053	gccatctgcgcagtaattagatcgcatcccggtatcaaacg	Amplifying homologous sequences to <i>viaA</i> gene
DJP054	atgaacagtggtgcgaaagcg	
DJP055	ccacaaggcatcgaacaagc	Assembling the upper three fragments through overlap PCR
DJP056	gtggaaccgggtactggaag	
DJP057	cttcttggtgctgtactca	RT-qPCR primer for <i>rpoA</i>
DJP058	tgggtgatcgcagcaagtg	
DJP059	ccgattgcaggatgagtt	RT-qPCR primer for <i>dnaA</i>
DJP060	taccaatcgaggacaaaac	
DJP061	cgttatcggaccatagaa	RT-qPCR primer for <i>gfp</i>
DJP062	cttcaaatctcactccgca	
DJP063	aagttaactgcgtggtact	RT-qPCR primer for <i>mcherry</i>
DJP064	acagggttcttgctttgta	
DJP065	tatgttgaaatttcgccc	RT-qPCR primer for <i>seqA</i>
DJP066	attcatcgaagcagaagt	
DJP067	gtattgacccgaacgaactg	RT-qPCR primer for <i>hns</i>
DJP068	agtcagggttttagtttcgc	

Table S5.

43 3'-TAMRA probes spanning the whole coding sequence of the *dnaA* gene

gccaaagcgaaagtgcacg	agaacgataggtcggttctg	cttcctgagatcgttcttta	cacgaagtcgatggatcg
tgcaatcgggcaagacactg	acgtgtgtttgacgtttacg	ggcgttgaaaggtgtgaaaa	tgaccagtttttctgcaat
aattctgtggctgtaactc	cgccagttggttagatttac	atctgttgattaccttcag	ccgtcttctgaatattgtcg
caatgggcgtatccacatac	ggaacaacgggttataggca	atagcgatccgaggtgagaa	cgcgactttgatctttagt
agcgtgtatcgcctagttc	atgcagcaggtgagttttac	caacgcccgttgatctcttc	atcgacgcttgaaaggaga
cccaatcaggacaaaaacgg	taaaccaacttggcattcgg	caaccgaagcgggatttcaa	tgtggttagtcagctcttc
ttattaaggtactgtcccg	tgaacaaagcgctcgagtg	ttttcatcaggatcgccac	caccaaagcatcgccaatc
gcagaaactggttagcagtc	tttgagggtttaaccatg	acgaatgtcgttttcgctcg	ttacggcaggcatgaagcac
cgacttcaaacgcagctgt	ttaactcttcgatcgctt	ggcgataaagaacgcactt	ctcttcacgcaactgctcga
gtagaaggcgcagcagttg	atctacggaacggtagtagc	gtacgttagatcgtagacgc	aatcttcttgatcgtgg
ggacgttatcccaacctgag	gaatatcgtcgatcagcagt	ggtaaagttggcattggcaa	

Parameter	Description	Values	Source
Titration model & Extrusion model			
λ	Biomass growth rate	$1.54h^{-1}$	This study (if not specifically defined)
α_A	DnaA synthesis rate	$300\ h^{-1}\mu m^{-3}$	Fitted in this study
$[A_f^c]$	Threshold of DnaA concentration	$10\ \mu m^{-3}$	Fitted in this study
α_H	Extruder synthesis rate	$180\ h^{-1}\mu m^{-3}$	Fitted in this study
η	Relative noise level of λ and α_A in stochastic model	0.1	<i>Fitted in this study</i>
Switch model			
$[D]_T$	Concentration of total DnaA protein	$400\mu m^{-3}$	(Berger & Wolde, 2022) SI table 2 LDDR model
α_l	Activation rate of DnaA-ATP by lipid	$750h^{-1}$	
α_{d1}	Activation rate of DnaA-ATP by the <i>DARS1</i> site	$100h^{-1}$	
α_{d2}	Activation rate of DnaA-ATP by the <i>DARS2</i> site	$643h^{-1}$	
β_{data}	Deactivation rate of DnaA-ATP by <i>data</i> site	$600h^{-1}$	
β_{rida}	Deactivation rate of DnaA-ATP by RIDA system	$500h^{-1}$	
K_D	Dissociation constant of DnaA activation and deactivation	$50\mu m^{-3}$	
K_D^P	Dissociation constant of DnaA promoter	$300\mu m^{-3}$	
f_C	Threshold of DnaA-ATP fraction for DNA replication initiation	0.75	
Titration switch model & Titration-switch-extrusion model			
K_D^P	Dissociation constant of DnaA promoter	$400\mu m^{-3}$	(Berger & Wolde, 2022) SI table switch-titration model
n	Cooperativity of dnaA expression	5	
α_H	Extruder synthesis rate	$550\ h^{-1}\mu m^{-3}$	Fitted in this study
$[A_{ATP}^{f_c}]$	Threshold of DnaA-ATP concentration for DNA replication initiation	$200\mu m^{-3}$	Fitted in this study

Table S6.

Parameters used in models

Acknowledgements

We thank numerous colleagues for discussions. We sincerely thank Dr. Aiguo Xia for his help with microscopy, and Dr. Sheng Yang for providing valuable suggestions on CRISPR-Cas9 system. Special thanks to Drs. Pan Chu and Yeqing Zong for kindly gifting us the CPP00458 and pPT plasmids. This research was financially supported by the National Natural Science Foundation of China (32025022, 32230062), Strategic Priority Research Program of the Chinese Academy of Sciences (XDB0480000), Joint NSFC-ISF Research Grant (32061143021) to C.L., US-Israel Binational Science Foundation (2017004) and Israel Science Foundation-NSCF joint research program (3320/20) to A.Z., National Natural Science Foundation of China (32170042), National Key R&D Program of China (2024YFA0916403), Youth Innovation Promotion Association CAS (number 2022369) to H.Z., National Key R&D Program of China (2021YFA0910703) to Y.B.; National Centre for Engineering Biology (NRF-MSG-2023-0003) and National University of Singapore (NUHSRO/2024/064/NUSMed/05/ SynCTI2.0) to M.W.C.

Additional information

Data availability

All data supporting the findings of this work are available from the corresponding author upon request. Source data are provided with this paper.

Code availability

Simulation data can be generated using the custom-made code and the parameter sets provided.

The code is publicly available at https://github.com/BaiYangBqdq/dynamics_of_biomass_DNA_coordination .

Author contributions

C.L. conceived the study and supervised all research. D.L. and T.W. designed the synthetic promoter library, validated reporters, and performed mRNA FISH experiments with assistance from H.Z. D.L., H.Z., and H.C. characterized H-NS phenotypes and conducted shift experiments. D.L. and X.H. executed DnaA shutdown experiments. C.L. and T.H. developed the extrusion model concept; Y.B., H.Z., and Z.Z. implemented mathematical modeling and simulations. A.Z. and M.W.C. contributed to data interpretation and manuscript refinement. All authors analyzed data, reviewed drafts, and approved the final manuscript. D.L., Y.B. and C.L. led manuscript writing.

Funding

National Natural Science Foundation of China (32025022)

National Natural Science Foundation of China (32230062)

National Natural Science Foundation of China (32170042)

Strategic Priority Research Program of the Chinese Academy of Sciences (XDB0480000)

Joint NSFC-ISF Research Grant (32061143021)

US-Israel Binational Science Foundation (2017004)

Israel Science Foundation-NSCF joint research program (3320/20)

National Key R&D Program of China (2024YFA0916403)

National Key R&D Program of China (2021YFA0910703)

Youth Innovation Promotion Association CAS (2022369)

National Centre for Engineering Biology (NRF-MSG-2023-0003)

National University of Singapore (NUHSRO/2024/064/NUSMed/05/ SynCTI2.0)

References

Balakrishnan R., Mori M., Segota I., Zhang Z., Aebersold R., Ludwig C., Hwa T. (2022) **Principles of gene regulation quantitatively connect DNA to RNA and proteins in bacteria** *Science* **378**:eabk2066 <https://doi.org/10.1126/science.abk2066> | Google Scholar

Berger M., Wolde P.R.t. (2022) **Robust replication initiation from coupled homeostatic mechanisms** *Nat. Commun* **13**:6556 <https://doi.org/10.1038/s41467-022-33886-6> | Google Scholar

Bintu L., Buchler N.E., Garcia H.G., Gerland U., Hwa T., Kondev J., Kuhlman T., Phillips R. (2005a) **Transcriptional regulation by the numbers: applications** *Curr. Opin. Genet. Dev* **15**:125–35 <https://doi.org/10.1016/j.gde.2005.02.006> | Google Scholar

Bintu L., Buchler N.E., Garcia H.G., Gerland U., Hwa T., Kondev J., Phillips R. (2005b) **Transcriptional regulation by the numbers: models** *Curr. Opin. Genet. Dev* **15**:116–24 <https://doi.org/10.1016/j.gde.2005.02.007> | Google Scholar

Boye E., Stokke T., Kleckner N., Skarstad K. (1996) **Coordinating DNA replication initiation with cell growth: Differential roles for DnaA and SeqA proteins** *Proc. Natl. Acad. Sci. U. S. A* **93**:12206–12211 <https://doi.org/10.1073/pnas.93.22.12206> | Google Scholar

Braun R.E., O'Day K., Wright A. (1985) **Autoregulation of the DNA replication gene dnaA in E. coli K-12** *Cell* **40**:159–169 [https://doi.org/10.1016/0092-8674\(85\)90319-8](https://doi.org/10.1016/0092-8674(85)90319-8) | Google Scholar

Bremer H., Churchward G., Young R. (1979) **Relation between growth and replication in bacteria** *J. Theor. Biol* **81**:533–45 [https://doi.org/10.1016/0022-5193\(79\)90051-1](https://doi.org/10.1016/0022-5193(79)90051-1) | Google Scholar

Campbell J.L., Kleckner N. (1990) **E. coli oriC and the dnaA gene promoter are sequestered from dam methyltransferase following the passage of the chromosomal replication fork** *Cell* **62**:967–79 [https://doi.org/10.1016/0092-8674\(90\)90271-f](https://doi.org/10.1016/0092-8674(90)90271-f) | Google Scholar

Christensen B.B., Atlung T., Hansen F.G. (1999) **DnaA boxes are important elements in setting the initiation mass of Escherichia coli** *J. Bacteriol* **181**:2683–2688 <https://doi.org/10.1128/jb.181.9.2683-2688.1999> | Google Scholar

- Dame R.T., Wyman C., Goosen N. (2000) **H-NS mediated compaction of DNA visualised by atomic force microscopy** *Nucleic Acids Res* **28**:3504–10 <https://doi.org/10.1093/nar/28.18.3504> | [Google Scholar](#)
- Dillon S.C., Dorman C.J. (2010) **Bacterial nucleoid-associated proteins, nucleoid structure and gene expression** *Nat. Rev. Microbiol* **8**:185–195 <https://doi.org/10.1038/nrmicro2261> | [Google Scholar](#)
- Donachie W.D., Blakely G.W. (2003) **Coupling the initiation of chromosome replication to cell size in Escherichia coli** *Curr. Opin. Microbiol* **6**:146–150 [https://doi.org/10.1016/s1369-5274\(03\)00026-2](https://doi.org/10.1016/s1369-5274(03)00026-2) | [Google Scholar](#)
- Dong M.J., Luo H., Gao F. (2023) **DoriC 12.0: an updated database of replication origins in both complete and draft prokaryotic genomes** *Nucleic Acids Res* **51**:D117–d120 <https://doi.org/10.1093/nar/gkac964> | [Google Scholar](#)
- Dorman C.J. (2004) **H-NS: a universal regulator for a dynamic genome** *Nat. Rev. Microbiol* **2**:391–400 <https://doi.org/10.1038/nrmicro883> | [Google Scholar](#)
- Ducret A., Quardokus E.M., Brun Y.V. (2016) **Microbej, a tool for high throughput bacterial cell detection and quantitative analysis** *Nat. Microbiol* **1**:16077 <https://doi.org/10.1038/nmicrobiol.2016.77> | [Google Scholar](#)
- Fragkos M., Ganier O., Coulombe P., Méchali M. (2015) **DNA replication origin activation in space and time** *Nat. Rev. Mol. Cell Biol* **16**:360–74 <https://doi.org/10.1038/nrm4002> | [Google Scholar](#)
- Fu H., Xiao F., Jun S. (2023) **Bacterial replication initiation as precision control by protein counting** *PRX Life* **1** <https://doi.org/10.1103/prxlife.1.013011> | [Google Scholar](#)
- Grimwade J.E., Leonard A.C. (2021) **Blocking, bending, and binding: regulation of initiation of chromosome replication during the Escherichia coli cell cycle by transcriptional modulators that interact with origin DNA** *Front. Microbiol* **12** <https://doi.org/10.3389/fmicb.2021.732270> | [Google Scholar](#)
- Hansen F.G., Atlung T. (2018) **The DnaA tale** *Front. Microbiol* **9** <https://doi.org/10.3389/fmicb.2018.00319> | [Google Scholar](#)
- Hansen F.G., Christensen B.B., Atlung T. (1991) **The initiator titration model: computer simulation of chromosome and minichromosome control** *Res. Microbiol* **142**:161–167 [https://doi.org/10.1016/0923-2508\(91\)90025-6](https://doi.org/10.1016/0923-2508(91)90025-6) | [Google Scholar](#)
- Hommais F., Krin E., Laurent-Winter C., Soutourina O., Malpertuy A., Le Caer J.P., Danchin A., Bertin P. (2001) **Large-scale monitoring of pleiotropic regulation of gene expression by the prokaryotic nucleoid-associated protein, H-NS** *Mol. Microbiol* **40**:20–36 <https://doi.org/10.1046/j.1365-2958.2001.02358.x> | [Google Scholar](#)
- Iuliani I., Mbemba G., Lagomarsino M.C., Sclavi B. (2024) **Direct single-cell observation of a key Escherichia coli cell-cycle oscillator** *Sci. Adv* **10**:ead5398 <https://doi.org/10.1126/sciadv.ado5398> | [Google Scholar](#)
- Jiang Y., Chen B., Duan C., Sun B., Yang J., Yang S. (2015) **Multigene editing in the Escherichia coli genome via the CRISPR-Cas9 system** *Appl. Environ. Microbiol* **81**:2506–2514 <https://doi.org/10.1128/aem.04023-14> | [Google Scholar](#)

- Kaguni J.M. (2011) **Replication initiation at the Escherichia coli chromosomal origin** *Curr. Opin. Chem. Biol* **15**:606–13 <https://doi.org/10.1016/j.cbpa.2011.07.016> | Google Scholar
- Katayama T., Kasho K., Kawakami H. (2017) **The DnaA cycle in Escherichia coli: activation, function and inactivation of the initiator protein** *Front. Microbiol* **8** <https://doi.org/10.3389/fmicb.2017.02496> | Google Scholar
- Khodursky A., Guzmán E.C., Hanawalt P.C. (2015) **Thymineless death lives on: new insights into a classic phenomenon** *Annu. Rev. Microbiol* **69**:247–263 <https://doi.org/10.1146/annurev-micro-092412-155749> | Google Scholar
- Knöppel A., Broström O., Gras K., Elf J., Fange D. (2023) **Regulatory elements coordinating initiation of chromosome replication to the Escherichia coli cell cycle** *Proc. Natl. Acad. Sci. U. S. A* **120**:e2213795120 <https://doi.org/10.1073/pnas.2213795120> | Google Scholar
- Løbner-Olesen A., Skarstad K., Hansen F.G., von Meyenburg K., Boye E. (1989) **The DnaA protein determines the initiation mass of Escherichia coli K-12** *Cell* **57**:881–9 [https://doi.org/10.1016/0092-8674\(89\)90802-7](https://doi.org/10.1016/0092-8674(89)90802-7) | Google Scholar
- Lu M., Campbell J.L., Boye E., Kleckner N. (1994) **SeqA: a negative modulator of replication initiation in E. coli** *Cell* **77**:413–26 [https://doi.org/10.1016/0092-8674\(94\)90156-2](https://doi.org/10.1016/0092-8674(94)90156-2) | Google Scholar
- Lutz R., Bujard H. (1997) **Independent and tight regulation of transcriptional units in Escherichia coli via the LacR/O, the TetR/O and AraC/I1-I2 regulatory elements** *Nucleic Acids Res* **25**:1203–10 <https://doi.org/10.1093/nar/25.6.1203> | Google Scholar
- Mäkelä J., Papagiannakis A., Lin W.H., Lanz M.C., Glenn S., Swaffer M., Marinov G.K., Skotheim J.M., Jacobs-Wagner C. (2024) **Genome concentration limits cell growth and modulates proteome composition in Escherichia coli** *eLife* **13** <https://doi.org/10.7554/eLife.97465> | Google Scholar
- Ohbayashi R., Hirooka S., Onuma R., Kanesaki Y., Hirose Y., Kobayashi Y., Fujiwara T., Furusawa C., Miyagishima S.-y. (2020) **Evolutionary changes in DnaA-dependent chromosomal replication in cyanobacteria** *Front. Microbiol* **11** <https://doi.org/10.3389/fmicb.2020.00786> | Google Scholar
- Ozaki S., Katayama T. (2012) **Highly organized DnaA-oriC complexes recruit the single-stranded DNA for replication initiation** *Nucleic Acids Res* **40**:1648–65 <https://doi.org/10.1093/nar/gkr832> | Google Scholar
- Pountain A.W., Jiang P., Yao T., Homae E., Guan Y., McDonald K.J.C., Podkowik M., Shopsin B., Torres V.J., Golding I., Yanai I. (2024) **Transcription-replication interactions reveal bacterial genome regulation** *Nature* **626**:661–669 <https://doi.org/10.1038/s41586-023-06974-w> | Google Scholar
- Saggioro C., Olliver A., Sclavi B. (2013) **Temperature-dependence of the DnaA-DNA interaction and its effect on the autoregulation of dnaA expression** *Biochem. J* **449**:333–341 <https://doi.org/10.1042/bj20120876> | Google Scholar
- Sekimizu K., Bramhill D., Kornberg A. (1987) **ATP activates DnaA protein in initiating replication of plasmids bearing the origin of the E. coli chromosome** *Cell* **50**:259–265 [https://doi.org/10.1016/0092-8674\(87\)90221-2](https://doi.org/10.1016/0092-8674(87)90221-2) | Google Scholar

- Si F., Le Treut G., Sauls J.T., Vadia S., Levin P.A., Jun S. (2019) **Mechanistic origin of cell-size control and homeostasis in bacteria** *Curr. Biol* **29**:1760–1770 <https://doi.org/10.1016/j.cub.2019.04.062> | Google Scholar
- Si F., Li D., Cox S.E., Sauls J.T., Azizi O., Sou C., Schwartz A.B., Erickstad M.J., Jun Y., Li X., Jun S. (2017) **Invariance of initiation mass and predictability of cell size in Escherichia coli** *Curr. Biol* **27**:1278–1287 <https://doi.org/10.1016/j.cub.2017.03.022> | Google Scholar
- Skinner S.O., Sepulveda L.A., Xu H., Golding I. (2013) **Measuring mRNA copy number in individual Escherichia coli cells using single-molecule fluorescent in situ hybridization** *Nat. Protoc* **8**:1100–1113 <https://doi.org/10.1038/nprot.2013.066> | Google Scholar
- Speck C., Weigel C., Messer W. (1999) **ATP- and ADP-DnaA protein, a molecular switch in gene regulation** *EMBO J* **18**:6169–6176 <https://doi.org/10.1093/emboj/18.21.6169> | Google Scholar
- Witz G., van Nimwegen E., Julou T. (2019) **Initiation of chromosome replication controls both division and replication cycles in E. coli through a double-adder mechanism** *eLife* **8** <https://doi.org/10.7554/eLife.48063> | Google Scholar
- Xia A., Zhang R., Huang Y., Ni L., Pu L., Li Y., Yang S., Jin F. (2022) **An adaptive tracking illumination system for optogenetic control of single bacterial cells** *Appl. Microbiol. Biotechnol* **106**:6775–6784 <https://doi.org/10.1007/s00253-022-12177-6> | Google Scholar
- Xie S., Swaffer M., Skotheim J.M. (2022) **Eukaryotic cell size control and its relation to biosynthesis and senescence** *Annu. Rev. Cell Dev. Biol* **38**:291–319 <https://doi.org/10.1146/annurev-cellbio-120219-040142> | Google Scholar
- Zaritsky A., Pritchard R.H. (1971) **Replication time of the chromosome in thymineless mutants of Escherichia coli** *J. Mol. Biol* **60**:65–74 [https://doi.org/10.1016/0022-2836\(71\)90447-5](https://doi.org/10.1016/0022-2836(71)90447-5) | Google Scholar
- Zheng H., Bai Y., Jiang M., Tokuyasu T.A., Huang X., Zhong F., Wu Y., Fu X., Kleckner N., Hwa T., Liu C. (2020) **General quantitative relations linking cell growth and the cell cycle in Escherichia coli** *Nat. Microbiol* **5**:995–1001 <https://doi.org/10.1038/s41564-020-0717-x> | Google Scholar
- Zheng H., Ho P.-Y., Jiang M., Tang B., Liu W., Li D., Yu X., Kleckner N.E., Amir A., Liu C. (2016) **Interrogating the Escherichia coli cell cycle by cell dimension perturbations** *Proc. Natl. Acad. Sci. U. S. A* **113**:15000–15005 <https://doi.org/10.1073/pnas.1617932114> | Google Scholar
- Zimmerman S.B. (2006) **Cooperative transitions of isolated Escherichia coli nucleoids: implications for the nucleoid as a cellular phase** *J. Struct. Biol* **153**:160–75 <https://doi.org/10.1016/j.jsb.2005.10.011> | Google Scholar
- Zong Y., Zhang H.M., Lyu C., Ji X., Hou J., Guo X., Ouyang Q., Lou C. (2017) **Insulated transcriptional elements enable precise design of genetic circuits** *Nat. Commun* **8**:52 <https://doi.org/10.1038/s41467-017-00063-z> | Google Scholar

Author information

Dengjin Li*

Key Laboratory of Quantitative Synthetic Biology, Shenzhen Institute of Synthetic Biology, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China, University of Chinese Academy of Sciences, Beijing, China
ORCID iD: [0009-0001-4148-5476](https://orcid.org/0009-0001-4148-5476)

*These authors contributed equally: Dengjin Li, Hai Zheng, Yang Bai.

Hai Zheng*

Key Laboratory of Quantitative Synthetic Biology, Shenzhen Institute of Synthetic Biology, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China, University of Chinese Academy of Sciences, Beijing, China

*These authors contributed equally: Dengjin Li, Hai Zheng, Yang Bai.

Yang Bai*

Key Laboratory of Quantitative Synthetic Biology, Shenzhen Institute of Synthetic Biology, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China, University of Chinese Academy of Sciences, Beijing, China
ORCID iD: [0000-0001-9976-2686](https://orcid.org/0000-0001-9976-2686)

*These authors contributed equally: Dengjin Li, Hai Zheng, Yang Bai.

Zheng Zhang

Key Laboratory of Quantitative Synthetic Biology, Shenzhen Institute of Synthetic Biology, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China

Hao Cheng

Key Laboratory of Quantitative Synthetic Biology, Shenzhen Institute of Synthetic Biology, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China

Xiongliang Huang

Key Laboratory of Quantitative Synthetic Biology, Shenzhen Institute of Synthetic Biology, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China, University of Chinese Academy of Sciences, Beijing, China

Ting Wei

Key Laboratory of Quantitative Synthetic Biology, Shenzhen Institute of Synthetic Biology, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China

Matthew Wook Chang

NUS Synthetic Biology for Clinical and Technological Innovation and Department of Biochemistry, Yong Loo Lin School of Medicine, National University of Singapore, Singapore, Singapore, National Centre for Engineering Biology, Singapore, Singapore

Arieh Zaritsky

Faculty of Natural Sciences, Ben-Gurion University of the Negev, Be'er Sheva, Israel

Terence Hwa

Department of Physics & Department of Molecular Biology, University of California, San Diego, La Jolla, United States
ORCID iD: [0000-0003-1837-6842](https://orcid.org/0000-0003-1837-6842)

Chenli Liu

Key Laboratory of Quantitative Synthetic Biology, Shenzhen Institute of Synthetic Biology, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China, University of Chinese Academy of Sciences, Beijing, China
ORCID iD: [0000-0003-3029-7207](https://orcid.org/0000-0003-3029-7207)

For correspondence: cl.liu@siat.ac.cn

Editors

Reviewing Editor

Ariel Amir

Weizmann Institute of Science, Rehovot, Israel

Senior Editor

Felix Campelo

Universitat Pompeu Fabra, Barcelona, Spain

Reviewer #1 (Public review):

Summary:

The study by Li and coworkers addresses the important and fundamental question of replication initiation in *Escherichia coli*, which remains open despite of many classic and recent works. It leverages single-cell mRNA-FISH experiments in strains with titratable DnaA and novel DnaA activity reporters to monitor DNA activity peaks versus size. The authors find oscillations in DnaA activity and show that their peaks correlate well with the estimated population-average replication initiation volume across conditions and imposed dnaA transcription levels. The study also proposes a novel and interesting extrusion model where DNA-binding proteins regulate free DnaA availability in response to biomass-DNA imbalance. Experimental perturbations of H-NS support the model validity, addressing key gaps in current replication control frameworks.

Strengths:

I find the study interesting and well conducted, and I think its main strong points are (i) the novel reporters obtained with systematic synthetic biology methods, and combined with a titratable dnaA strain, (ii) the interesting perturbations (titration, production arrest and H-NS) and (iii) the use of single-cell mRNA FISH to monitor transcripts directly. The proposed extrusion model is also interesting, though not fully validated, and I think it will contribute positively to the future debate.

Weaknesses and Limitations

A relevant limitation in novelty is that DnaA activity and concentration oscillations have been reported by the cited Iuliani and coworkers previously by dynamic microscopy, and to a smaller extent by the other cited study by Pountain and coworkers using mRNA FISH.

An important limitation is that the study is not dynamic. While monitoring mRNA is interesting and relevant, the current study is based on concentrations and not time variations (or nascent mRNA). Conversely, the study by Iuliani and coworkers, while having the drawback of monitoring proteins it can access directly production rates. It would be interesting for future studies to monitor the strains and reporters dynamically, as well as using (as a control) the technique of this study on the chromosomal reporters used by Iuliani et al.

While the implemented code is made available and the parameter values are given in the text, important details are missing regarding the mathematical models (mathematical definitions, clear discussions of ingredients and main assumptions, and choices made in the deployment of such models, which are presented briefly in the Methods section). The reader is not given sufficient tools to understand the predictions of different models and no analytical estimates are used and the falsification procedures are not clear. More transparency and depth in the analysis would be needed to use the models as more than a heuristic tool for qualitative arguments. The Berger model for example has many parameters and many regimes and behaviors. When models are compared to data (e.g. in fig. 2G) it is not clear how parameters were fixed, and whether and how the model prediction depends on adjustable parameters.

Importantly, the statement about tight correlations of peak volumes and average estimated initiation volume does not establish coincidence. Crucially, the data rely on average initiation volumes, and the estimate procedure relies on assumptions that could lead to systematic biases and uncertainties added to the population variability (in any case error bars are not provided).

The delays observed by the authors (in both directions) between the peaks of DnaA-activity conditional averages with respect to volume and the average estimated initiation volumes are not incompatible with those observed dynamically by Iuliani and coworkers. The direct experiment to prove the authors' point would be to use a direct proxy of replication initiation such as SeqA or DnaN and monitor initiations and quantify DnaA activity peaks jointly, with dynamic measurements.

While not being an expert I had the doubt that the fact that the reporters are on plasmid (despite a normalization control that seems very sensible) might affect the measurements. The approach is different from the aforementioned previous study, which used a chromosomal reporter placed symmetrically, at the same distance from the origin of replication as the original dnaA promoter.

Overall Appraisal:

In summary, this appears to me as a very interesting study providing valuable high-precision data and a novel testable hypothesis, the extrusion model, supported by relevant perturbation experiments and open to future explorations.

Comments on revisions:

I am happy with the replies and the revisions.

The main outstanding point remains that reconstructing the mathematical model details from the text (and having to rely on the code) is not optimal for a reader. However, I do understand that the authors intend to use the models as a heuristic tool only and possibly plan a theoretical study where they explore the models more systematically.

<https://doi.org/10.7554/eLife.107214.2.sa2>

Reviewer #2 (Public review):

Summary:

The authors show that in *E. coli* the initiator protein DnaA oscillates post-translationally: its activity rises and peaks exactly when DNA replication begins, even if *dnaA* transcription is held constant. To explain this, they propose an "extrusion" mechanism in which nucleoid-associated proteins such as H-NS, whose amount grows with cell volume, dislodge DnaA from chromosomal binding sites; modelling and H-NS perturbations reproduce the observed drop in initiation mass and extra initiations seen after *dnaA* shut-down. Together, the data and model link biomass growth to replication timing through chromosome-driven, post-translational control of DnaA, filling gaps left by classic titration and ATP/ADP-switch models.

Strengths:

- (1) Introduces an "extrusion" model that adds a new post-translational layer to replication control and explains data unexplained by classic titration or ATP/ADP-switch frameworks.
- (2) A major asset of the study is that it bridges the longstanding gap between DnaA oscillations and DNA-replication initiation, providing direct single-cell evidence that pulses of DnaA activity peak exactly at the moment of initiation across multiple growth conditions and genetic perturbations.
- (3) A tunable *dnaA* strain and targeted H-NS manipulations shift initiation mass exactly as the model predicts, giving model-driven validation across growth conditions.
- (4) A purpose-built Psyn66 reporter combined with mRNA-FISH captures DnaA-activity pulses with cell-cycle resolution, providing direct, compelling data.

Weaknesses:

- (1) What happens to the (C+D) period and initiation time as the *dnaA* mRNA level changes? This is not discussed in the text or figure and should be addressed.
- (2) It is unclear what is meant by "relative *dnaA* mRNA level." Relative to what? Wild-type expression? Maximum expression? This should be explicitly defined.
- (3) It would be helpful to provide some intuition for why an increase in *dnaA* mRNA level leads to a decrease in initiation mass per ori and an increase in oriC copy number.
- (4) The titration and switch models do not explicitly include *dnaA* mRNA in the dynamics of DnaA protein. Yet, in Figure 2G, initiation mass is shown to decrease linearly with *dnaA* mRNA level in these models. How was *dnaA* mRNA level represented or approximated in these simulations?
- (5) Is Schaechter's law (i.e., exponential scaling of average cell size with growth rate) still valid under the different *dnaA* mRNA expression conditions tested?
- (6) The manuscript should explain more explicitly how the extrusion model implements post-translational control of DnaA and, in particular, how this yields the nonlinear drop in relative initiation mass versus *dnaA* mRNA seen in Fig. 6E. Please provide the governing equation that links total DnaA, the volume-dependent "extruder" pool, and the threshold of free DnaA at initiation, and show-briefly but quantitatively-how this equation produces the observed concave curve.
- (7) Does this Extrusion model give well well-known adder per origin, i.e., initiation to initiation is an adder.

(8) DnaA protein or activity is never measured; mRNA is treated as a linear proxy. Yet the authors' own narrative stresses post-translational (not transcriptional) control of DnaA. Without parallel immunoblots or activity readouts, it is impossible to know whether a six-fold mRNA increase truly yields a proportional rise in active DnaA.

(9) Figure 2 infers both initiation mass and oriC copy number from bulk measurements (OD₆₀₀ per cell and rifampicin-cephalexin run-out) instead of measuring them directly in single cells. Any DnaA-dependent changes in cell size, shape, or antibiotic permeability could skew these bulk proxies, so the plotted relationships may not accurately reflect true initiation events.

Comments on revisions:

The authors have addressed all of my previous concerns, questions, and suggestions sufficiently.

<https://doi.org/10.7554/eLife.107214.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

The study by Li and coworkers addresses the important and fundamental question of replication initiation in Escherichia coli, which remains open, despite many classic and recent works. It leverages single-cell mRNA-FISH experiments in strains with titratable DnaA and novel DnaA activity reporters to monitor DNA activity peaks versus size. The authors find oscillations in DnaA activity and show that their peaks correlate well with the estimated population-average replication initiation volume across conditions and imposed dnaA transcription levels. The study also proposes a novel extrusion model where DNA-binding proteins regulate free DnaA availability in response to biomass-DNA imbalance. Experimental perturbations of H-NS support the model validity, addressing key gaps in current replication control frameworks.

Strengths:

I find the study interesting and well conducted, and I think its main strong points are:

(1) the novel reporters obtained with systematic synthetic biology methods, and combined with a titratable dnaA strain.

(2) the interesting perturbations (titration, production arrest, and H-NS).

(3) the use of single-cell mRNA FISH to monitor transcripts directly.

The proposed extrusion model is also interesting, though not fully validated, and I think it will contribute positively to the future debate.

We thank the reviewer for acknowledging the strengths of our study.

Weaknesses and Limitations:

(1) A relevant limitation in novelty is that DnaA activity and concentration oscillations have been reported by the cited Iuliani and coworkers previously by dynamic microscopy, and to a smaller extent by the other cited study by Pountain and coworkers using mRNA FISH.

(2) An important limitation is that the study is not dynamic. While monitoring mRNA is interesting and relevant, the current study is based on concentrations and not time variations (or nascent mRNA). Conversely, the study by Iuliani and coworkers, while having the drawback of monitoring proteins, can directly assess production rates. It would be interesting for future studies or revisions to monitor the strains and reporters dynamically, as well as using (as a control) the technique of this study on the chromosomal reporters used by Iuliani et al.

We acknowledge the value of dynamic measurements and clarify our methodological rationale.

While Iuliani et al. provided valuable temporal resolution through protein dynamics, our mRNA FISH approach achieves direct decoupling of transcriptional vs. post-translational regulation (Fig 4F-H), and condition flexibility across 7 growth rates (30-66 min doubling times). This trade-off sacrifices temporal resolution for enhanced population-scale resolution and perturbation flexibility. To directly address temporal coupling, future work will implement dual-color live imaging of DnaA activity concurrent with replication initiation events.

(3) Regarding the mathematical models, a lot of details are missing regarding the definitions and the use of such models, which are only presented briefly in the Methods section. The reader is not given any tools to understand the predictions of different models, and no analytical estimates are used. The falsification procedures are not clear. More transparency and depth in the analysis are needed, unless the models are just used as a heuristic tool for qualitative arguments (but this would weaken the claims). The Berger model, for example, has many parameters and many regimes and behaviors. When models are compared to data (e.g., in Figure 2G), it is not clear which parameters were used, how they were fixed, and whether and how the model prediction depends on parameters.

We agree that model transparency is essential for quantitative validation. To address this, all model parameters (DnaA synthesis rate, activation/deactivation rates etc.) are explicitly tabulated in Supplementary Information Table S6. For the titration (Hansen et al. 1991) and extrusion models, we derive analytical expressions for initiation mass (IM) sensitivity to DnaA expression in Supplementary Note 1. For Figure 2G/S6, we used published parameters (Berger & Wolde 2022 SI Table 2) with experiment growth conditions ($\mu = 1.54 \text{ h}^{-1}$).

The extrusion model's validation relies primarily on its ability to resolve paradoxical initiation events under *dnaA* shutdown (Fig 6C), a test where other models fail categorically. While the Berger titration-switch hybrid can fit steady-state IM trends (Fig S6A), it cannot reproduce post-shutdown dynamics without ad hoc modifications (Fig S6B). We acknowledge that comprehensive analysis of all model regimes exceeds this study's scope but provide full simulation code for independent verification: https://github.com/BaiYangBqdq/dynamics_of_biomass_DNA_coordination

(4) Importantly, the main statement about tight correlations of peak volumes and average estimated initiation volume does not establish coincidence, and some of the claims by the authors are unclear in these respects (e.g., when they say "we resolve a 1:1 coupling between DnaA activity thresholds and replication initiation", the statement

could be correct but is ambiguous). Crucially, the data rely on average initiation volumes (on which there seems to be an eternally open debate, also involving the authors), and the estimate procedure relies on assumptions that could lead to biases and uncertainties added to the population variability (in any case, error bars are not provided).

We acknowledge the limitations of population-level inference and have refined our claims: "Replication initiation volume scales proportionally with peak DnaA activity volume with a slope of 1.0 ($R_2=0.98$, Fig 7G), indicating predictive correspondence rather than absolute coincidence. While population-level V_i estimation cannot resolve single-cell stochasticity, the consistent $V^*: V_i$ relationship across 20 conditions suggest DnaA activity thresholds predict initiation timing within physiological error margins". Future work will implement simultaneously DnaA activity and replication forks by using microfluidic single-cell tracking.

(5) The delays observed by the authors (in both directions) between the peaks of DnaA activity conditional averages with respect to volume and the average estimated initiation volumes are not incompatible with those observed dynamically by Iuliani and coworkers. The direct experiment to prove the authors' point would be to use a direct proxy of replication initiation, such as SeqA or DnaN, and monitor initiations and quantify DnaA activity peaks jointly, with dynamic measurements.

We acknowledge the observed temporal deviations between DnaA activity peaks (V^*) and population-derived volumes at initiation (V_i) in certain conditions, in line with the findings of Iuliani et al. This might be mechanistically consistent with the time required for orisome assembly or *oriC* sequestration. They do not contradict our core finding that initiation occurs at a defined DnaA activity threshold (slope=1.0, $R_2=0.98$ in $V^*: V_i$ correlation).

(6) While not being an expert, I had some doubt that the fact that the reporters are on plasmid (despite a normalization control that seems very sensible) might affect the measurements. Also, I did not understand how the authors validated the assumptions that the reporters are sensitive to DnaA-ATP specifically. It seems this assumption is validated by previous studies only.

We employed a plasmid-based reporter system to circumvent the significant confounding effects of chromosomal position on promoter activity, as extensively documented by Pountain et al., where local genomic context (e.g., nucleoid occlusion, supercoiling gradients, and neighboring operons) introduces uncontrolled variability. By housing the P_{syn66} test promoter and P_{con} normalization control in identical low-copy pSC101 vectors (<8 copies/cell, Peterson & Phillips, Plasmid 2008), we ensured they experience equivalent physical and biochemical environments. This ratiometric design, where DnaA activity is calculated, actively corrects for global fluctuations in RNA polymerase availability, nucleotide pools, and plasmid copy number. Critically, P_{syn66} 's architecture emulates natural DnaA-responsive elements: its strong DnaA boxes report free DnaA concentration, while its weak box is preferentially bound by DnaA-ATP (Speck et al., EMBO journal 1999), mirroring the nucleotide-state sensitivity of *oriC* and the native *dnaA* promoter. This system was indispensable for our central finding, as it uniquely enabled the decoupling of DnaA activity oscillations from transcriptional feedback (Fig. 4F-H), an experiment fundamentally impossible with chromosomally integrated reporters due to autoregulatory interference.

Overall Appraisal:

In summary, this appears as a very interesting study, providing valuable data and a novel hypothesis, the extrusion model, open to future explorations. However, given several limitations, some of the claims appear overstated. Finally, the text contains some

selfevaluations, such as "our findings redefine the paradigm for replication control", etc., that appear exaggerated.

We thank the reviewer for highlighting the need for precise language in framing our conclusions. We have implemented the following substantive revisions throughout the manuscript to ensure claims align strictly with empirical evidence:

- (1) Changed "redefine the paradigm for replication control" into "advance the paradigm for replication control" (Introduction)
- (2) Changed "redefine bacterial cell cycle control" into "refine bacterial cell cycle control as a dynamic interplay..." (Discussion)
- (3) Removed the term "spatial" from the Discussion's description of DnaA-chromosome interactions (Discussion, first paragraph).
- (4) Changed "provides a blueprint" into "provides a valuable tool for dissecting spatial regulation..." (Discussion, final paragraph)
- (5) Scrutinized all superlatives (e.g., "critical feat" into "important capability"; "fundamental principle of cellular organization" into "potential organizational strategy")
- (6) Replaced the instances of "robust" with evidence-backed descriptors (e.g., "sensitive," "consistent")
- (7) We agree that the extrusion model requires further validation and have emphasized this in Discussion: "While H-NS perturbation supports extrusion mechanism, future work should identify the full extruder interactome and elucidate how metabolic signals modulate their activity" (final paragraph)

This calibrated language more accurately represents our study as a conceptual advance with testable mechanisms, not a complete paradigm shift.

Reviewer #2 (Public review):

Summary:

The authors show that in E. coli, the initiator protein DnaA oscillates post-translationally: its activity rises and peaks exactly when DNA replication begins, even if dnaA transcription is held constant. To explain this, they propose an "extrusion" mechanism in which nucleoid-associated proteins such as H-NS, whose amount grows with cell volume, dislodge DnaA from chromosomal binding sites; modelling and H-NS perturbations reproduce the observed drop in initiation mass and extra initiations seen after dnaA shut-down. Together, the data and model link biomass growth to replication timing through chromosome-driven, posttranslational control of DnaA, filling gaps left by classic titration and ATP/ADP-switch models.

Strengths:

- (1) Introduces an "extrusion" model that adds a new post-translational layer to replication control and explains data unexplained by classic titration or ATP/ADP-switch frameworks.*
- (2) A major asset of the study is that it bridges the longstanding gap between DnaA oscillations and DNA-replication initiation, providing direct single-cell evidence that pulses of DnaA activity peak exactly at the moment of initiation across multiple growth conditions and genetic perturbations.*

(3) A tunable *dnaA* strain and targeted H-NS manipulations shift initiation mass exactly as the model predicts, giving model-driven validation across growth conditions.

(4) A purpose-built *Psyn66* reporter combined with mRNA-FISH captures DnaA-activity pulses with cell-cycle resolution, providing direct, compelling data.

We thank the reviewer for acknowledging the strengths of our study.

Weaknesses:

(1) What happens to the (C+D) period and initiation time as the *dnaA* mRNA level changes? This is not discussed in the text or figure and should be addressed.

We thank the reviewer for this important observation. Our data demonstrate that increased *dnaA* mRNA levels induce two compensatory changes in cell cycle progression:

(1) Earlier replication initiation, manifested as a reduced initiation mass: the initiation mass decreased from 5.6 to 2.6 ($\text{OD}_{600}\cdot\text{ml}$ per 10^{10} cells) as the relative *dnaA* mRNA level increased from 0.2 to 7.2 (normalized to the wild-type level) (Fig. 2F, red).

(2) Prolonged C+D period: Increased by approximately 60% (from 1.05 to 1.66 hours, Fig. 2F blue).

The complete quantitative relationship is now explicitly described in the Results section: “Concurrently, the initiation mass was reduced by 50%, and the period from initiation to division (C+D) was increased by ~60% (Fig. 2F)”

(2) It is unclear what is meant by “relative *dnaA* mRNA level.” Relative to what? Wild-type expression? Maximum expression? This should be explicitly defined.

The relative *dnaA* mRNA level was obtained by normalizing to that in wild-type MG1655 cells grown in the same medium. To clarify this point, we have now marked the wild-type level in Fig. 1B, and a clear description of this has also been included in the figure caption.

(3) It would be helpful to provide some intuition for why an increase in *dnaA* mRNA level leads to a decrease in initiation mass per *ori* and an increase in *oriC* copy number.

Thank you for your valuable suggestion. Increased *dnaA* mRNA accelerates DnaA accumulation, causing cells to reach the initiation threshold at a smaller cell size (reducing initiation mass, Fig. 2F red). This earlier initiation increases *oriC* copies per cell at populational level (Fig. 2E). This mechanistic interpretation now appears in the Results: “As the DnaA expression level increases, DnaA activity reaches the initiation threshold earlier. Given that cell mass remained nearly unchanged, this earlier initiation led to an increase in population-averaged cellular *oriC* numbers (Fig. 2E).”

(4) The titration and switch models do not explicitly include *dnaA* mRNA in the dynamics of DnaA protein. Yet, in Figure 2G, initiation mass is shown to decrease linearly with *dnaA* mRNA level in these models. How was *dnaA* mRNA level represented or approximated in these simulations?

All models presented in this article omit explicit modeling of *dnaA* mRNA dynamics for simplicity. However, at steady state, the relative level of *dnaA* mRNA can be approximated by the relative expression rate of DnaA protein, as both reflect the expression level of DnaA. This detail is now clarified in the caption of Figure 2G.

(5) Is Schaechter's law (i.e., exponential scaling of average cell size with growth rate) still valid under the different *dnaA* mRNA expression conditions tested?

Schaechter's law describes the exponential scaling of average cell size with growth rate in bacteria. In our prior work (Zheng et al., Nature Microbiology 2020), where we demonstrated that Schaechter's law fails in slow-growth regimes. However, in current study, growth rate remained constant across different *dnaA* expression levels (Fig. 2C), and cell mass showed no significant change (Fig. 2D). Since Schaechter's law specifically addresses how cell size scales with growth rate, it does not apply here, as growth rate was invariant in our perturbations, which selectively alter replication initiation dynamics, not growth rate or size scaling.

(6) The manuscript should explain more explicitly how the extrusion model implements posttranslational control of DnaA and, in particular, how this yields the nonlinear drop in relative initiation mass versus *dnaA* mRNA seen in Figure 6E. Please provide the governing equation that links total DnaA, the volume-dependent "extruder" pool, and the threshold of free DnaA at initiation, and show - briefly but quantitatively - how this equation produces the observed concave curve.

The governing equations linking initiation mass and DnaA expression level is now provided in Supplementary Note S1 for both the titration and the extrusion model. In general, the dependence of initiation mass (V_I) on *dnaA* expression level (α_A) dependency takes an inverse

1 proportionality form: $V_I \propto \frac{1}{\alpha_A}$. In the extrusion model, the incorporated extruder protein is

assumed to have similar synthesis dynamics as DnaA and can release DnaA from DnaA-box. After denoting the synthesis rate of the extruder as α_H , the combined effect of DnaA and the

extruder on replication initiation can be briefly described as: $V_I \propto \frac{1}{\alpha_A + \alpha_H}$. Then the additive

contribution of α_H dampens the sensitivity of initiation mass to changes in α_A , resulting in a significantly flattened curve. As a result, the predicted $V_I - \alpha_A$ relationship has a concave shape in the semi-log plots.

(7) Does this Extrusion model give well well-known adder per origin, i.e., initiation to initiation is an adder.

Yes, the extrusion model can provide the initiation-to-initiation adder phenomenon, this information was provided in fig. S3C.

(8) DnaA protein or activity is never measured; mRNA is treated as a linear proxy. Yet the authors' own narrative stresses post-translational (not transcriptional) control of DnaA. Without parallel immunoblots or activity readouts, it is impossible to know whether a sixfold mRNA increase truly yields a proportional rise in active DnaA.

We acknowledge the reviewer's valid concern regarding the indirect nature of our DnaA activity measurements. While mRNA levels alone cannot resolve active DnaA dynamics, our approach integrates functional replication outcomes with a validated synthetic reporter to infer activity. Crucially, elevated *dnaA* mRNA causes demonstrable biological effects: earlier replication initiation (Fig. 2F) and increased *oriC* copies (Fig. 2E), directly confirming enhanced functional DnaA activity at the *oriC* locus. The P_{syn66} reporter, engineered with DnaA-boxes mirroring *oriC*'s architecture, provides orthogonal validation, showing

progressive repression to *dnaA* induction (Fig. 3C). Our operational metric $k_{\text{syn66}} = \frac{[m2](P_{\text{syn66}})}{[m2](P_{\text{con}})}$, bases

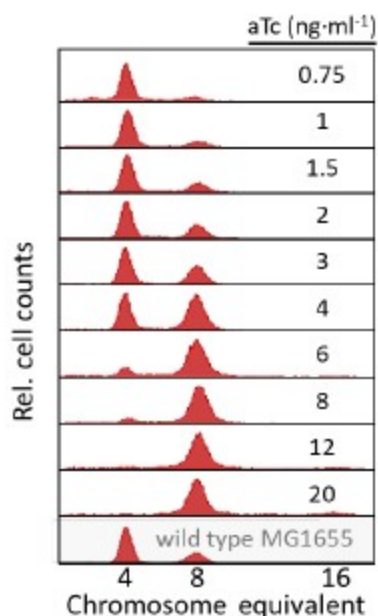
on P_{syn66} responds sensitively to DnaA-chromosome interactions within its characterized 8-fold dynamic range (Fig. 3C). Immunoblots would be inadequate here, as they cannot distinguish functionally critical pools: free versus chromosome-bound DnaA, or DnaA-ATP versus DnaAADP, precisely the post-translational states our study implicates in regulation. We therefore prioritize functional readouts (initiation timing) and the P_{syn66} reporter, which probes the biologically active fraction relevant to replication control.

(9) Figure 2 infers both initiation mass and *oriC* copy number from bulk measurements (OD_{600} per cell and rifampicin-cephalexin run-out) instead of measuring them directly in single cells. Any DnaA-dependent changes in cell size, shape, or antibiotic permeability could skew these bulk proxies, so the plotted relationships may not accurately reflect true initiation events.

We acknowledge the reviewer's valid methodological concern and clarify that while bulk measurements carry inherent limitations, our approach is grounded in established techniques with demonstrated reliability. Cell mass was inferred from OD_{600}/cell , which correlates strongly with direct dry weight measurements and microscopic cell volumes across diverse growth conditions, as validated in our prior work (Zheng et al., Nature Microbiology 2020). Crucially, cell mass remained invariant across *dnaA* expression levels (Fig. 2D).

Regarding *oriC* quantification, the rifampicin-cephalexin run-out assay is a widely applied for replication initiation studies. Our data shows expected 2^n *oriC* distributions without abnormal ploidy (as shown below). While single-cell methods offer superior resolution, our bulk approach provides accurate population-level trends.

Author response image 1.



Recommendations for the authors:

Reviewing Editor Comments:

The reviewers felt that the mathematical modeling was not adequately explained in the paper, and that this affected the readability of the manuscript. The authors are encouraged to elaborate on this aspect of the paper (in addition to strengthening other claims, if possible, per the reviewers' comments).

We thank the editor and reviewers for their constructive feedback. We have comprehensively strengthened the mathematical modeling framework to enhance clarity and rigor.

Reviewer #1 (Recommendations for the authors):

The only revision I would do is a recalibration of the claims and a major effort to clarify the modeling part (including a detailed SI appendix), without necessarily performing additional work.

To enhance mathematical modeling transparency, we have completed model description in the method section and a parameter table with literature-sourced values in Supplementary Information Table S6. Moreover, analytical derivations of initiation mass dependencies are performed and presented in the Supplementary Information Note S1.

Of course, there are extra experiments (mentioned in the public review) that would help support some of the big claims, but that can be considered a different project.

Thank you for your suggestion. This will be addressed in our future work.

Minor suggestion: please put signposts or plot jointly to compare the maxima/minima in Figures 4D, E, G, and H.

We added dashed lines in Figures 4D, and E, to synchronize visualization of DnaA activity peaks and transcriptional minima across panels, facilitating direct biological comparisons.

Reviewer #2 (Recommendations for the authors):

(1) Should define what DNA activity is.

We have explicitly defined DnaA activity in the Introduction as “the capacity to initiate replication...” and noted that it is “governed by free DnaA concentration, DnaA-ATP/-ADP ratio, and orisome assembly competence”.

(2) Word repetition - “...grown in in Luria-Bertani (LB) medium...”.

Corrected.

(3) Typographical error - “FISH ... was preformed” should be “performed”.

Corrected.

(4) The manuscript alternates between “ng ml⁻¹” and “ng·ml⁻¹”; choose one style and apply it uniformly.

Standardized the units to ng·ml⁻¹ throughout.

(5) *Reference duplicates* - Some citations appear twice in the bibliography (e.g., "Bintu et al., 2005a/b" and "Bintu et al., 2005b" listed again later).

The studies by Bintu et al. (2005a, 2005b) represent separate works: 2005a details applications, and 2005b develops models.

<https://doi.org/10.7554/eLife.107214.2.sa0>